

Dimension structure in simplex stability

Entry 005 research manuscript

AI-assisted manuscript candidate

8 October 2026

Abstract

We study approximation of a convex body by each prescribed maximum-volume inscribed simplex, with dilation centered at that simplex's own centroid, using the projection-body pyramid defect. For polytopes with at most $d + 2$ facets, the optimal coefficient of the sharp power $1/(d - 1)$ has asymptotic value d : its ratio to d tends to one. The geometric basis is an exact classification of all maximum simplices in a simplex cut by one halfspace, including nonvertex maxima, through weighted rooted forests. Signed maximal cofactors of the actual facet area-normal/support matrix give an exact formula for the defect. We also derive concentric cores and a face-depth hierarchy from full projection control, and prove linear-dimensional retention for edge-supported configurations through inverse-minor rigidity of stochastic matrices with two-supported columns. We reproduce the finite actual-facet enclosure argument and preserve its compatibility conditions. The unrestricted gap between linear lower and quadratic upper dimension orders remains open.

Status. This is an unpublished manuscript candidate. The new results have written proofs and the source-specific traditional review record described in Appendix C; they have not been formalized in Lean. Independent model review, parent integration review, and an author's self-reviewed contribution are distinguished there. No professional peer-review, novelty certification, or priority claim is made.

Contents

1	The original invariant and the dimension question	2
2	All maximum simplices in a single halfspace clipping	3
3	The actual defect from signed facet cofactors	6
4	The asymptotically optimal coefficient for few facets	8
5	Structure forced by all directional projections	10
6	Inverse minor rigidity in structured classes	16
A	A finite determinant assignment theorem	26
B	From the actual facet law to an enclosing simplex	27
C	Proof coverage and verification boundaries	30

1 The original invariant and the dimension question

For a full-dimensional convex body $K \subset \mathbb{R}^d$, write $|K|$ for its Euclidean volume and h_K for its support function. For a unit vector u , let $\pi_K(u)$ denote the $(d-1)$ -dimensional volume of the orthogonal projection onto $u^\perp$. Its homogeneous extension is

$$b_K(z) = \begin{cases} \|z\| \pi_K(z/\|z\|), & z \neq 0, \\ 0, & z = 0. \end{cases}$$

The canonical projection body ΠK has support function b_K . For a body L of dimension p , put

$$R(L) = \frac{|\Pi L|}{|L|^{p-1}}, \quad \text{pyr } L = \text{conv}((L, 0) \cup \{(0, 1)\}).$$

The actual projection-body pyramid quantities are

$$a(K) = \left(\frac{d}{d+1}\right)^d \frac{R(\text{pyr } K)}{R(K)} - 1, \quad e(K) = a(K) - \frac{1}{d+1}. \quad (1.1)$$

For a full-dimensional simplex $S \subset K$ with its own centroid g_S , we define the excess

$$E(K, S) = \inf\{u \geq 0 : K \subset g_S + (1+u)(S - g_S)\}. \quad (1.2)$$

The simplex S is prescribed. In particular, the definition does not minimize over maximum simplices or over centers. All affine changes below transport K , S , and g_S together. Set

$$n = d+1, \quad m = d-1, \quad H_d = 3d^2(d+1)^3(d+2).$$

Unless stated otherwise $d \geq 3$. The geometric clipping results also apply in smaller dimensions within their explicitly stated ranges.

The retained unrestricted theorem [3] gives $E(K, S) \leq 4096d^2e(K)^{1/m}$ for every prescribed maximum simplex; realized square-pyramid bodies give a lower coefficient of $n[dn]^{1/m}$. Its proof and formal-verification record are prior inputs, not formal verification of the statements added here. Our principal result settles the leading dimensional coefficient on an entire facet-count class.

Theorem 1.1 (The few-facet coefficient). *For every full-dimensional convex d -polytope K with at most $d+2$ facets and every maximum-volume inscribed simplex S ,*

$$E(K, S) \leq nH_d^{1/m}e(K)^{1/m} \leq 192de(K)^{1/m}. \quad (1.3)$$

The power $1/m$ is sharp in this class. If

$$G_d^{\text{fac}} = \sup \left\{ \frac{E(K, S)}{e(K)^{1/m}} : K \text{ has at most } d+2 \text{ facets, } S \text{ is maximum, } e(K) > 0 \right\},$$

where all bodies are full dimensional, then

$$n[dn]^{1/m} \leq G_d^{\text{fac}} \leq nH_d^{1/m}, \quad \lim_{d \rightarrow \infty} \frac{G_d^{\text{fac}}}{d} = 1. \quad (1.4)$$

The exact fixed-dimensional optimum is not determined. The proof in Section 4 uses two exact descriptions: all maxima in an arbitrary one-halfspace clipping, and the actual defect from signed facet cofactors. The halfspace may remove several vertices of its original simplex.

Full projection control has additional content beyond near-corner points. Section 5 develops inscribed concentric cores and a complete face-depth hierarchy. The resulting

comparison simplex is not identified with a prescribed maximum. Section 6 proves inverse rigidity and retention when columns have support at most two; its actual-defect conclusion requires compatibility with the enclosing simplex supplied by the actual chain. These conclusions leave the unrestricted linear-versus-quadratic dimension-order question open.

The actual facet identities, finite determinant assignment, and enclosure proof are reproduced in Appendices A–B. Their standard convex-geometric inputs are Cauchy’s projection formula, the zonotope volume formula, the first mixed-volume formula, and Minkowski’s first inequality; Brunn–Minkowski is also used for one of the core proofs. We use these classical results in their ordinary forms [1]. A systematic literature and priority comparison has not been carried out. Source-series provenance [2] and the precise review boundaries are recorded separately in Appendix C and the accompanying source map.

2 All maximum simplices in a single halfspace clipping

Theorem 2.1 (Exact clipping retention). *Let P be a full-dimensional d -simplex, let H be a closed halfspace, and suppose $K = P \cap H$ is full dimensional. With $n = d + 1$ and the P -barycentric coordinates λ_i , put*

$$\rho_* = \max_i \left(1 - \max_{x \in K} \lambda_i(x) \right).$$

Every prescribed maximum-volume simplex $S \subset K$ satisfies

$$E(K, S) \leq n\rho_*. \quad (2.1)$$

Consequently corner losses at most $\rho \geq 0$ imply $E(K, S) \leq n\rho$. The coefficient n is sharp for every $d \geq 2$ and every $0 < \rho_ < 1$. All maximum simplices and their exact excess are described below.*

2.1 The vertices and a candidate maximum

Use affine P -coordinates, so $P = \text{conv}(e_1, \dots, e_n)$ in $\sum_i x_i = 1$. Write $H = \{f \geq 0\}$ and $f_i = f(e_i)$. If no f_i is negative, $K = P$ and the result is immediate. Otherwise full dimension implies that at least one f_i is positive. Partition the indices as

$$I = \{i : f_i = -a_i < 0\}, \quad J = \{j : f_j = b_j > 0\}, \quad Z = \{z : f_z = 0\}.$$

Let $B = \max_{j \in J} b_j > 0$, $J_* = \{j : b_j = B\}$, and

$$t_i = \frac{B}{a_i + B}, \quad s_i = \frac{a_i}{a_i + B} = 1 - t_i \quad (i \in I).$$

The vertices of K are precisely

$$e_j \ (j \in J \cup Z), \quad v_{ij} = \frac{b_j e_i + a_i e_j}{a_i + b_j} \quad (i \in I, \ j \in J). \quad (2.2)$$

Indeed, a new vertex lies on an old edge crossing the cut hyperplane. Equivalently, the active coordinate constraints leave at most two positive coordinates, and the two-coordinate case gives this expression.

For $z \in Z$, only the vertex e_z has a nonzero z -coordinate, so it is mandatory in every full-dimensional vertex simplex. Its column can be expanded out. The determinant calculation

may therefore be performed on $I \cup J$ and all e_z then adjoined. Fix $j_* \in J_*$. The simplex with every retained e_j and with v_{ij_*} for every $i \in I$ has normalized determinant

$$D_* = \prod_{i \in I} t_i > 0. \quad (2.3)$$

Expansion in the retained standard columns proves this identity. Moreover,

$$\max_K \lambda_i = t_i \ (i \in I), \quad \max_K \lambda_j = 1 \ (j \in J \cup Z), \quad \rho_* = \max_{i \in I} s_i. \quad (2.4)$$

2.2 Weighted rooted forests and equality

Select $|I| + |J|$ columns from the vertices with these row indices. Represent each v_{ij} by an edge ij in the bipartite graph on $I \cup J$ and each selected e_j by a root mark at j . Include isolated row vertices. Multiply row $i \in I$ by a_i and row $j \in J$ by $-b_j$. The edge and root columns become, respectively,

$$\frac{a_i b_j}{a_i + b_j} (e_i - e_j), \quad -b_j e_j.$$

Columns in a connected component form a basis of its coordinate space exactly when that component is a tree with one selected root. An unrooted component on v vertices has incidence rank $v - 1$. A rooted component has rank at most v , and full independence together with the total column count requires exactly v columns in each component. Connectivity needs at least $v - 1$ edges and full rank needs at least one root, hence there are exactly $v - 1$ edges and one root. Cycles or excess roots force column dependence. Conversely, leaf elimination shows that an unweighted rooted tree has determinant of absolute value one.

Orient every edge of a rooted forest toward its root. Leaf elimination before row scaling gives the exact product

$$|\det W| = \prod_{\text{nonroot vertices } u} (\text{entry at } u \text{ of its parent-edge column}). \quad (2.5)$$

A bad vertex $i \in I$ is never a root; if its parent is j , its factor $b_j/(a_i + b_j)$ is at most t_i . A nonroot good vertex has factor $a_i/(a_i + b_j) < 1$. Thus every selected determinant satisfies

$$|\det W| \leq \prod_{i \in I} t_i = D_*.$$

Equality holds exactly when every good vertex is a root and every bad vertex attaches directly to a member of J_* . All vertex maxima are therefore

$$\text{conv}(\{e_j : j \in J \cup Z\} \cup \{v_{ij_*} : i \in I\}), \quad j_* \in J_*, \quad (2.6)$$

with independent choices of j_i . Ordered by their original bad and retained indices, all these oriented determinants equal D_* . Expanding an arbitrary simplex's determinant separately in convex combinations of K -vertices proves that its absolute determinant is also at most D_* .

2.3 Every prescribed maximum including nonvertex maxima

Let S be any maximum and express each of its vertices as a convex combination of vertices of K . Sample the combinations independently. The oriented determinant of S is the expectation of the sampled oriented determinant. Equality in

$$|\mathbb{E} \det W| \leq \mathbb{E} |\det W| \leq D_*$$

forces every positive-probability tuple to be a maximum from (2.6), with the same orientation.

If no column selected a mandatory retained vertex e_j surely, independence would give positive probability that the whole tuple omitted it. Hence each retained vertex occupies a deterministic column. After removing those columns, each of the disjoint groups $G_i = \{v_{ij} : j \in J_*\}$ must appear exactly once in every tuple. The same avoidance argument forces some column to belong surely to each G_i . Distinct groups require distinct columns, accounting for all remaining columns. Merely relabeling the vertices of this same S gives

$$\begin{aligned} S &= \text{conv}(\{e_j : j \in J \cup Z\} \cup \{w_i : i \in I\}), \\ w_i &= t_i e_i + s_i z_i, \quad z_i \in \text{conv}\{e_j : j \in J_*\}. \end{aligned} \quad (2.7)$$

Conversely every such simplex has determinant D_* and is maximum. This includes all continuous tie cases.

2.4 Exact excess about the same centroid

Write $z_i = \sum_{j \in J_*} z_{ji} e_j$, with $z_{ji} \geq 0$ and $\sum_j z_{ji} = 1$. The S -coordinates of an affine point x are

$$\begin{aligned} \alpha_i(x) &= (1 + a_i/B)x_i \quad (i \in I), \\ \alpha_j(x) &= x_j - \sum_{i \in I} (a_i/B) z_{ji} x_i \quad (j \in J_*), \\ \alpha_j(x) &= x_j \quad (j \notin I \cup J_*). \end{aligned} \quad (2.8)$$

For $j \in J_*$ put $B_{-j} = \max(\{b_h : h \in J, h \neq j\} \cup \{0\})$. The exact largest negative coordinate magnitude is

$$\beta(S) = \max_{\substack{i \in I \\ j \in J_*}} \frac{z_{ji} a_i B_{-j}}{B(a_i + B_{-j})}. \quad (2.9)$$

A zero B_{-j} contributes zero. At v_{ih} with $h \neq j$, the possibly negative coordinate is $-z_{ji} a_i b_h / [B(a_i + b_h)]$. At v_{ij} it is $s_i(1 - z_{ji}) \geq 0$, and retained vertices have nonnegative coordinates. Monotonicity of $b/(a_i + b)$ gives (2.9); convex averaging covers every $x \in K$. Since $B_{-j} \leq B$, it follows that $\beta(S) \leq \max_i a_i / (a_i + B) = \rho_*$.

A centroid dilation by $1 + u$ has all simplex coordinates at least $-u/n$. Consequently the exact identity is

$$E(K, S) = n\beta(S) \leq n\rho_*. \quad (2.10)$$

This proves Theorem 2.1 for the prescribed simplex and its original centroid. If $|J_*| \geq 2$, then $\beta(S) = \max_i (s_i \max_{j \in J_*} z_{ji})$. Every vertex maximum has $E = n\rho_*$, while balanced nonvertex maxima can have smaller excess. If $J_* = \{j_*\}$, the maximum is unique as a set, and

$$\beta(S) = \max_i \frac{a_i B_2}{B(a_i + B_2)},$$

where B_2 is the second-largest positive value, or zero if there is none. If $|J| = 1$, K itself is a simplex and $\beta = 0$. An equal-intercept one-vertex truncation has one bad vertex and, for $d \geq 2$, at least two tied good vertices. Setting $s_i = \rho$ and taking a vertex maximum gives $\beta = \rho$, proving the stated sharpness. The graph in this proof concerns clipping vertices; it is not the facet-normal spanning-two-forest construction for a $d + 2$ -vertex circuit.

2.5 An arbitrary body containing an opposite facet

Theorem 2.2 (One-corner bodies). *Let $P = \text{conv}(v_0, \dots, v_d)$ be full dimensional and let K be a convex body with $\text{conv}(v_1, \dots, v_d) \subset K \subset P$. Put $t = \max_K \lambda_0 > 0$ and $\rho = 1 - t$. Every*

prescribed maximum simplex is

$$S = \text{conv}(v_1, \dots, v_d, q), \quad q \in K, \quad \lambda_0(q) = t,$$

and $E(K, S) \leq (d+1)\rho$. The coefficient is sharp for $d \geq 2$ and $0 < \rho < 1$. No polyhedral or boundary-regularity hypothesis is imposed.

Proof. If $t = 1$, the maximizing point is v_0 , so $K = P$. Otherwise set $L = \{x \in P : \lambda_0(x) \leq t\}$. By the complete clipping classification, its maximum volume is $t|P|$, and all maxima consist of the opposite facet and an apex at height t . Compactness gives an apex $q \in K$ attaining that height, and the entire opposite facet is in K . This simplex has volume $t|P|$, so K and L have the same maximum volume. Every prescribed maximum of K is therefore a maximum of L and has precisely the claimed form. Its apex belongs to K , without any requirement that it be an extreme point of K .

Write the coordinates of q as $(t, q_1, \dots, q_d)$, where $q_j \geq 0$ and $\sum_j q_j = \rho$. For $x \in K$, its S -coordinates are

$$\alpha_0(x) = \lambda_0(x)/t, \quad \alpha_j(x) = \lambda_j(x) - (q_j/t)\lambda_0(x) \geq -q_j \geq -\rho.$$

The centroid formula proves the stronger estimate $E(K, S) \leq (d+1) \max_j q_j$. Equality in the asserted bound is attained by L and an apex on one truncated edge, using a second truncated edge to realize the negative coordinate. $\square$

If the actual enclosure chain supplies a P satisfying the additional opposite-facet containment hypothesis, the theorem may be used with $\rho = (H_d e(K))^{1/m}$. That extra hypothesis is not asserted for an arbitrary body or an unrelated enclosing simplex.

3 The actual defect from signed facet cofactors

Let K be a full-dimensional d -polytope with exactly $N = d+2$ irredundant facets. Put $n = d+1$. For facet i , let ν_i be its outward unit normal, $s_i > 0$ its actual area, and h_i its support number. Set

$$g_i = s_i \nu_i, \quad U_i = (g_i, s_i h_i) \in \mathbb{R}^{d+1}, \quad V = |K| > 0.$$

The divergence identities give

$$\sum_i g_i = 0, \quad \sum_i s_i h_i = dV, \quad \sum_i U_i = (0, \dots, 0, dV). \quad (3.1)$$

The horizontal rows of U have rank d because the facet normals span $\mathbb{R}^d$. The last column-sum identity supplies an independent vertical direction, so U has rank n . Define the signed maximal cofactors

$$z_i = (-1)^{i-1} \det(U_1, \dots, U_{i-1}, U_{i+1}, \dots, U_N). \quad (3.2)$$

Then $z \neq 0$ and $\sum_i z_i U_i = 0$. Translating the origin to the interior of K leaves these determinants invariant by a row operation and makes all $h_i > 0$. Thus $\sum_i z_i s_i h_i = 0$ forces both positive and negative entries in z .

Theorem 3.1 (Exact cofactor defect). *With $L = \sum_i |z_i|$ and $T = \sum_{i < j} |z_i - z_j|$, the original invariant satisfies*

$$a(K) = \frac{L}{T}, \quad \delta(K) := n - \frac{1}{a(K)} = n - \frac{T}{L}, \quad e(K) = \frac{\delta(K)}{n(n - \delta(K))}. \quad (3.3)$$

In particular,

$$\delta(K) = \frac{2}{L} \sum_{\substack{i < j \\ z_i z_j > 0}} \min(|z_i|, |z_j|). \quad (3.4)$$

Proof. Let $H = \sum_{i < j} |\det(g_k : k \notin \{i, j\})|$, with the remaining columns in increasing order. The actual finite-facet identity proved in Appendix B.1 gives $a(K) = L/(dVH)$. We show

$$dV |\det(g_k : k \notin \{i, j\})| = |z_i - z_j|. \quad (3.5)$$

Fix $i < j$. In the n -column matrix omitting column i , replace its column U_j by $\sum_k U_k$. By multilinearity only the original term and the term replacing U_j by U_i survive. Reordering the latter columns with the signs in (3.2) gives $z_i - z_j$ up to an overall sign. The replacement column is $(0, \dots, 0, dV)$ by (3.1), and expansion along it gives the other side of (3.5), again up to sign. Hence $dVH = T$ and $a = L/T$. Solving $e = a - 1/n$ proves (3.3); all its denominators are positive.

For real x, y , the difference $|x| + |y| - |x - y|$ equals $2 \min(|x|, |y|)$ when $xy > 0$, and equals zero otherwise. Each $|z_i|$ occurs in n unordered pairs. Summing this identity gives (3.4). $\square$

3.1 Signed concentration

Sort the positive entries as $p_1 \geq \dots \geq p_p > 0$ and the absolute values of the negative entries as $q_1 \geq \dots \geq q_q > 0$. Then

$$\delta(K) = \frac{2}{L} \left(\sum_{i=2}^p (i-1)p_i + \sum_{j=2}^q (j-1)q_j \right). \quad (3.6)$$

For $\varepsilon = (L - p_1 - q_1)/L$ this gives

$$2\varepsilon \leq \delta(K) \leq 2 \max(p-1, q-1)\varepsilon \leq 2d\varepsilon. \quad (3.7)$$

Indeed each omitted weight in (3.6) has coefficient between one and $\max(p-1, q-1)$, and both sign classes are nonempty. At least $1 - \delta(K)/2$ of the normalized absolute cofactor mass is therefore concentrated on one positive and one negative entry. The weighted identity, rather than only this comparison, determines the defect. Common nonzero rescaling and overall sign reversal leave all formulas unchanged. Zero cofactors, including pyramid degeneracies, are permitted; no simplicity or genericity condition is required. Arbitrary signed vectors are not asserted to be realizable facet data.

3.2 Explicit clipping coordinates

Suppose $K = P \cap \{f \geq 0\}$ with $f = \sum_i f_i \lambda_i$, and now assume that all n old facets and the cut facet remain irredundant. Use the partition $f_i = -a_i < 0$ on I , $f_j = b_j > 0$ on J , allowing zero values. For any origin there are unique positive scales r_i, q such that

$$U_i = r_i(-\nabla \lambda_i, \lambda_i(0)), \quad U_c = q(-\nabla f, f(0)).$$

Since $\sum_i \lambda_i = 1$, the column-sum identity and independence of the affine barycentric functions imply

$$r_i + qf_i = R, \quad R = dV > 0, \quad (3.8)$$

so each $r_i = R - qf_i$ is positive. The affine identity for f gives the kernel vector, up to common scale,

$$z_c = 1, \quad z_i = -qf_i/r_i.$$

Write

$$\alpha_i = \frac{qa_i}{R + qa_i} \in (0, 1), \quad \beta_j = \frac{qb_j}{R - qb_j} > 0.$$

The cut entry and each α_i have the same sign, with $\min(1, \alpha_i) = \alpha_i$, so (3.4) becomes

$$\delta(K) = \frac{2 \left[\sum_i \alpha_i + \sum_{i < i'} \min(\alpha_i, \alpha_{i'}) + \sum_{j < j'} \min(\beta_j, \beta_{j'}) \right]}{1 + \sum_i \alpha_i + \sum_j \beta_j}. \quad (3.9)$$

Zero f_i give zero cofactors. If clipping removes an entire old facet, that redundant facet must be deleted before applying this calculation; one may not divide by a vanished area.

For the equal-intercept truncation

$$K_t = \{x_i \geq 0 : t \leq \sum_i x_i \leq 1\}, \quad 0 < t < 1, \quad (3.10)$$

the normalized cofactor vector is

$$z_c = 1, \quad z_0 = t^d, \quad z_1 = \dots = z_d = -\beta, \quad \beta = \frac{t^{d-1}(1-t)}{1-t^{d-1}}.$$

The old opposite facet retains its whole area, each other old facet retains the fraction $1 - t^{d-1}$, and the cut area scale is the fraction t^{d-1} . These facts, or substitution in (3.8), give the displayed vector. Therefore

$$\delta(K_t) = \frac{2[t^d + \binom{d}{2}\beta]}{1 + t^d + d\beta}. \quad (3.11)$$

Substitution into (3.3) yields the exact original defect and its fixed-dimensional leading term

$$\begin{aligned} e(K_t) &= \frac{t^{d-1}[d(d-1) - (d+1)(d-2)t - 2t^d]}{(d+1)(1-t^d)(d+1 + (d-1)t^{d-1})}, \\ e(K_t) &= \frac{d(d-1)}{(d+1)^2} t^{d-1} + O(t^d). \end{aligned} \quad (3.12)$$

For a square followed by $d-2$ pyramids, the four nonzero cofactors are proportional to $(1, -1, 1, -1)$ and the remaining $d-2$ vanish. Thus $\delta = 1$ and $e = 1/[d(d+1)]$. The direct facet certificate in the next section checks the same actual invariant without assuming realizability of a merely formal cofactor vector.

4 The asymptotically optimal coefficient for few facets

4.1 From the actual facet law to a single clipping

Proposition B.1, proved in full in the appendices, gives the precise finite-facet interface needed here: for $0 \leq e(K) \leq H_d^{-1}$ there is an enclosing simplex P , formed by exactly $d+1$ distinct irredundant K -facet halfspaces, such that

$$\pi_P(u) - \pi_K(u) \leq H_d e(K) \pi_P(u) \quad (\|u\| = 1). \quad (4.1)$$

It uses the actual atoms $X_F = \nu_F/h_F$ and probabilities $p_F = s_F h_F/(d|K|)$. In particular, the selected halfspaces are actual facets, rather than unspecified support planes inferred from a compact limit. The determinant identity is exactly $(B-A)/B = ne/(1+ne)$ for this law. The assignment bound is $h \leq n^3(n+1)e/(1+ne)$; at the upper endpoint $h \leq 1/(3d^2) < 1/(2d)$, so its strict enclosure gate remains valid.

We also record the projection-cap argument explicitly.

Lemma 4.1 (Near corners from all projections). *If $K \subset P$, with P a full-dimensional simplex, and $\pi_P(u) - \pi_K(u) \leq \delta \pi_P(u)$ for every unit u , where $0 \leq \delta \leq 1$, then each corner loss satisfies*

$$1 - \max_{x \in K} \lambda_i^P(x) \leq \delta^{1/(d-1)}.$$

Proof. Let q_i be the corresponding vertex and let $t_i = 1 - \max_K \lambda_i^P \in [0, 1]$. Choose a unit direction parallel to its opposite facet. The affine function λ_i^P descends under this projection, and has range $[0, 1]$ on the projected simplex. For $0 < s < t_i$, the projected homothetic cap $q_i + s(P - q_i)$ is contained in $\{\lambda_i^P \geq 1 - s\}$ and disjoint from the projection of K . Its projected volume is $s^{d-1} \pi_P(u)$. Both sets lie in the projection of P , so $s^{d-1} \leq \delta$. Letting $s \uparrow t_i$ proves the claim; $t_i = 0$ is immediate. Strict caps avoid any assumption about the measure of the boundary, and include $\delta = 0$. $\square$

Proof of the upper bound in Theorem 1.1. A full-dimensional polytope has at least $d+1$ facets. With exactly $d+1$ it is a simplex, and $E = e = 0$. Otherwise it has $d+2$ facets. The P above uses $d+1$ of them, so $K = P \cap H$ for the remaining facet halfspace. Set $\rho = (H_d e(K))^{1/m}$. Lemma 4.1 bounds all corner losses by ρ , and Theorem 2.1 gives $E(K, S) \leq n\rho$ for this same originally prescribed maximum. The argument includes $e = 0$ and $e = H_d^{-1}$. For larger e , the universal maximum-simplex bound $E \leq n$ suffices, since $\rho > 1$.

For the uniform scalar estimate, $H_3 = 8640 \leq 144^2$, and $H_{d+1}/H_d \leq 18 \leq 64$ follows from $(d+1)/d \leq 2$, $(d+2)/(d+1) \leq 3/2$, and $(d+3)/(d+2) \leq 4/3$ in its squared, cubed, and linear factors. Induction gives $H_d \leq 144^{d-1}$. Since $n \leq 4d/3$, $nH_d^{1/m} \leq 192d$. $\square$

4.2 Sharp exponent and a realized lower coefficient

The body K_t in (3.10) has exactly $d+2$ facets. Theorem 2.1 identifies vertex maxima with $E(K_t, S_t) = nt$, and (3.12) gives $e(K_t) \sim d(d-1)t^m/n^2$. Thus for every fixed d and every $\eta > 1/m$, the ratio E/e^η diverges as $t \downarrow 0$. No such stronger exponent can hold uniformly even with a dimension-dependent coefficient.

For the all-scale lower coefficient take

$$C_d = \text{conv}(0, e_1, e_2, e_1 + e_2, e_3, \dots, e_d), \quad S_d = \text{conv}(0, e_1, \dots, e_d).$$

This is a square followed by $d-2$ pyramids, with irredundant inequalities

$$x_i \geq 0 \ (1 \leq i \leq d), \quad x_1 + \sum_{i=3}^d x_i \leq 1, \quad x_2 + \sum_{i=3}^d x_i \leq 1.$$

Every full-dimensional vertex simplex contains all pyramid vertices and three of the four square vertices. Every such square triangle has area $1/2$, so the resulting volume is $1/d!$. Separate determinant expansion in convex combinations bounds arbitrary inscribed simplices by the same volume. Thus S_d is maximum. The omitted square vertex has coordinates $(-1, 1, 1, 0, \dots, 0)$ in S_d , so $E(C_d, S_d) = n$.

For completeness, a direct actual-facet certificate gives the defect. Set $c = 1/(d-1)!$. The area-normal/support pairs are

$$(-ce_1, 0), \quad (-ce_2, 0), \quad (-2ce_i, 0) \ (3 \leq i \leq d),$$

$$\left(c \left(e_1 + \sum_{i=3}^d e_i \right), c \right), \quad \left(c \left(e_2 + \sum_{i=3}^d e_i \right), c \right).$$

The volume is $2/d!$. For the raw sums of Appendix B.1, direct minors give

$$\mathcal{H}(C_d) = c^d d 2^{d-1}, \quad \mathcal{L}(C_d) = c^{d+1} 2^d.$$

Only four lifted subsets contribute, each by $c^{d+1} 2^{d-2}$: omit one top facet or one of the first two coordinate facets. For horizontal minors the contributions with zero, one, and two top facets

are respectively $1, d, d-1$, multiplied by $c^d 2^{d-2}$. The exact actual identity $a = \mathcal{L}/(d|C_d|\mathcal{H})$ therefore gives $a(C_d) = 1/d$ and $e(C_d) = 1/(dn)$. Hence $G_d^{\text{fac}} \geq n[dn]^{1/m}$.

Finally, H_d is a fixed polynomial in d , so $H_d^{1/m} \rightarrow 1$; also $[dn]^{1/m} \rightarrow 1$. Divide the two coefficient bounds by d and squeeze to obtain (1.4). No uniformity in a fixed-dimensional truncation remainder is needed.

5 Structure forced by all directional projections

Let $P = \text{conv}(p_0, \dots, p_d) \subset \mathbb{R}^d$ be a full-dimensional simplex, let $K \subset P$ be a convex body, and assume throughout this section that

$$\pi_K(v) \geq (1 - \delta)\pi_P(v) \quad \text{for every unit direction } v, \quad 0 \leq \delta \leq 1, \quad d \geq 3. \quad (5.1)$$

Here $\pi_L(v) = \text{Vol}_{d-1}(\text{proj}_{v^\perp} L)$ denotes actual Euclidean projection volume for unit v . Directions constructed below are normalized when applying this hypothesis. Write

$$n = d + 1, \quad m = d - 1, \quad \rho = \delta^{1/(d-1)}, \quad c_P = \frac{1}{n} \sum_{i=0}^d p_i.$$

We first retain an elementary concentric-core argument and its Brunn–Minkowski refinement, and then prove a stronger finite-spectrum bound. These conclusions use the complete hypothesis (5.1), rather than only the existence of one point close to each vertex. They construct a comparison simplex with centroid c_P ; this simplex is not asserted to be maximum-volume and need not coincide with an originally prescribed maximum simplex S .

5.1 Centroid caps and a separating projection

Lemma 5.1 (Elementary centroid cap). *Let D be an m -dimensional simplex with centroid g . Every closed halfspace H whose boundary contains g satisfies*

$$\text{Vol}_m(D \cap H) \geq \frac{\text{Vol}_m(D)}{m+1}. \quad (5.2)$$

Consequently, if $z \in D$, $0 \leq \theta \leq 1$, and $y = (1 - \theta)z + \theta g$, every closed halfspace H_y through y satisfies

$$\text{Vol}_m(D \cap H_y) \geq \frac{\theta^m}{m+1} \text{Vol}_m(D). \quad (5.3)$$

Proof. Use barycentric coordinates $x_0, \dots, x_m$ on D and write $H = \{L \geq 0\}$ with $L(g) = 0$. Let σ cyclically permute these coordinates. Each σ^k preserves D and its volume, while

$$\sum_{k=0}^m L(\sigma^k x) = (m+1)L(g) = 0.$$

At least one summand is nonnegative. The $m+1$ inverse cyclic images of $D \cap H$ therefore cover D , proving (5.2). A constant functional causes no difficulty. For the second assertion, take the parallel centroid halfspace and apply the homothety $x \mapsto (1 - \theta)z + \theta x$. Its image lies in $D \cap H_y$ and has the volume asserted in (5.3). $\square$

Lemma 5.2 (A separator-parallel simplicial projection). *Let L be a nonconstant affine functional on P . There is a nonzero direction v parallel to the level hyperplanes of L for which $D = \text{proj}_{v^\perp} P$ is an $(d-1)$ -simplex. If g is the centroid of D , then*

$$c'_P = \frac{d}{n}g + \frac{1}{n}r' \quad (5.4)$$

for a point $r' \in D$, where primes denote this orthogonal projection. The functional L descends to a nonconstant affine functional on D .

Proof. Choose distinct indices a, b such that $L(p_a)$ and $L(p_b)$ are, respectively, the minimum and maximum vertex values. Choose any $j \notin \{a, b\}$. A point $r \in [p_a, p_b]$ satisfies $L(r) = L(p_j)$. Put $v = p_j - r$. This vector is nonzero and parallel to the level hyperplanes of L . Orthogonal projection along v sends p_j to r' . The other d projected vertices are affinely independent: otherwise v would be parallel to the facet opposite p_j , whereas $p_j - r$ is transverse to that facet. Their convex hull is D , because r' lies on their projected edge $[p'_a, p'_b]$. Summing the projected vertices gives (5.4). The two extreme values of L are distinct and survive the projection, so the descended functional is nonconstant. Repeated extreme values or equality of $L(p_j)$ with an endpoint value do not affect the construction. The construction itself requires only $d \geq 2$. $\square$

5.2 Two concentric-core bounds

Theorem 5.3 (Elementary concentric core). *Define*

$$\kappa_d = \frac{d+1}{d} d^{1/(d-1)}.$$

If $t = \kappa_d \rho \leq 1$, then

$$(1-t)P + tc_P \subset K. \quad (5.5)$$

Moreover, $\kappa_d \leq 4\sqrt{3}/3 < 7/3$ for all $d \geq 3$.

Proof. Fix $0 < t \leq 1$ and suppose that $y = (1-t)x + tc_P \notin K$ for some $x \in P$. Strict separation gives a nonconstant affine functional L with $\max_K L < L(y)$. Apply Lemma 5.2, so that the projection is parallel to the levels of L . By (5.4),

$$\begin{aligned} y' &= (1-t)x' + \frac{t}{n}r' + \frac{td}{n}g = (1-\theta)z + \theta g, \\ \theta &= \frac{td}{n}, \quad z = \frac{(1-t)x' + (t/n)r'}{1 - td/n} \in D. \end{aligned}$$

The denominator is positive even for $t = 1$. Since $\theta > 0$, y' is interior to D in its affine hull. The projection of K lies strictly on the lower- L side of y' . The opposite cap has volume at least

$$\frac{1}{d} \left(\frac{td}{n} \right)^{d-1} \text{Vol}_{d-1}(D)$$

by Lemma 5.1. In fact, the missing volume is strictly larger: the strip between $\max_K L$ and $L(y')$ has positive $(d-1)$ -volume because y' is interior. Hence

$$\delta > \frac{1}{d} \left(\frac{td}{n} \right)^{d-1}. \quad (5.6)$$

For $t = \kappa_d \rho > 0$, the right-hand side is $\rho^{d-1} = \delta$, a contradiction. This also proves the $t = 1$ endpoint. If $\delta = 0$, the same contradiction applies for every $t > 0$; letting t decrease to zero and using closedness of K gives $K = P$.

Finally, $d \leq 3^{(d-1)/2}$ for $d \geq 3$. There is equality at $d = 3$, and the ratio $(d+1)/d$ is at most $4/3 < \sqrt{3}$ in the inductive step. Thus $d^{1/(d-1)} \leq \sqrt{3}$, and the stated uniform bound follows from $(d+1)/d \leq 4/3$. $\square$

The preceding proof is independent of Brunn–Minkowski. For completeness we also retain the stronger centroid-cap argument, including the one-dimensional derivation used in the refinement.

Lemma 5.4 (Centroid cap from Brunn–Minkowski). *Let D be an m -dimensional convex body with centroid g . Every closed halfspace H through g satisfies*

$$\frac{\text{Vol}_m(D \cap H)}{\text{Vol}_m(D)} \geq \left(\frac{m}{m+1} \right)^m. \quad (5.7)$$

Proof. Let X be uniform in D and let $Z = L(X)$, where L is the affine functional defining H , oriented so that $H = \{L \geq 0\}$ and normalized by $\mathbb{E}Z = 0$. A constant functional is immediate, so assume L is nonconstant. Set

$$F(u) = \mathbb{P}(Z \geq u), \quad R(u) = F(u)^{1/m}.$$

For $0 \leq \alpha \leq 1$,

$$(1 - \alpha)(D \cap \{L \geq u\}) + \alpha(D \cap \{L \geq v\}) \subset D \cap \{L \geq (1 - \alpha)u + \alpha v\}.$$

Brunn–Minkowski therefore makes R concave. Between the minimum and maximum of L on D , this function is continuous and strictly decreasing. Its inverse T on $(0, 1)$ is also concave and decreasing: apply concavity of R to $T(s_0), T(s_1)$ and then use its decreasing monotonicity. The distribution of Z is continuous, and the probability-integral transform makes $F(Z)$ uniform on $(0, 1)$. Consequently $U = R(Z)$ has density ms^{m-1} and

$$\mathbb{E}U = \frac{m}{m+1}, \quad \mathbb{E}T(U) = \mathbb{E}Z = 0.$$

Jensen’s inequality gives $T(m/(m+1)) \geq 0$. Since T is decreasing, $R(0) \geq m/(m+1)$, proving (5.7). These expectations are bounded one-dimensional integrals; equivalently, the Jensen step follows by integrating a supporting line for T at $m/(m+1)$. $\square$

Theorem 5.5 (Sharper concentric core). *The conclusion of Theorem 5.3 holds with*

$$t = \frac{d+1}{d-1} \rho \leq 1. \quad (5.8)$$

In particular, the coefficient $(d+1)/(d-1)$ is at most 2.

Proof. Repeat the proof of Theorem 5.3, replacing the centroid fraction $1/d$ by $((d-1)/d)^{d-1}$ from Lemma 5.4. The same homothety and strictly positive strip show that a missing core point would force

$$\delta > \left(\frac{t(d-1)}{d+1} \right)^{d-1}.$$

The choice (5.8) contradicts this inequality. Strict separation again includes $t = 1$, and the closed limit used above includes $\delta = 0$. $\square$

5.3 An elementary finite-spectrum refinement

Theorem 5.6 (Finite-spectrum concentric core). *Put*

$$M_0 = 1, \quad M_k = \frac{k^k}{(k+1)^{2k+1}} \quad (k \geq 1), \quad \tau_d(\delta) = n \max_{1 \leq \ell \leq m} (M_{\ell-1} \delta)^{1/\ell}.$$

If $0 \leq \tau_d(\delta) \leq 1$, then

$$(1 - \tau_d(\delta))P + \tau_d(\delta)c_P \subset K. \quad (5.9)$$

This conclusion uses neither Brunn–Minkowski nor a centroid-cap inequality. One may use the smallest parameter among the applicable core bounds.

Proof. We first obtain an explicit missing-cap estimate. Let $b \geq 0$ be affine on P , normalized so that its minimum vertex value is zero and the sum of all its vertex values is one. List its other d vertex values as

$$B \geq C \geq b_1 \geq \dots \geq b_{d-2} \geq 0.$$

Let $0 < s \leq 1/n$ and assume K lies strictly in $\{b > s\}$. Project along the line joining the vertex of value C to the point of value C on the edge with endpoint values 0 and B . As in Lemma 5.2, this direction is parallel to the levels of b and gives an m -simplex D whose vertex values are

$$0, B, b_1, \dots, b_{d-2}.$$

The value- C vertex has become redundant. Since $B \geq 1/d > s$, the cap $\{b \leq s\}$ contains the simplex with apex at the zero-valued vertex and edge-intercept fractions $\min(1, s/a)$ for each other endpoint value a , with fraction one when $a = 0$. Its relative volume is exactly

$$R_s = \frac{s^m}{B \prod_{i=1}^{d-2} \max(b_i, s)}. \quad (5.10)$$

All of it is missing from the projection of K . The actual missing volume is strictly larger, since K lies strictly above level s and D has positive-volume strips immediately above that level.

Let k be the number of b_i exceeding s , so $0 \leq k \leq m-1$. For $k = 0$ the denominator in (5.10) is at most $s^{m-1} = M_0 s^{m-1}$. For $k \geq 1$, let U be the sum of these k active values. As C is at least each active value, $C \geq U/k$, and normalization gives

$$1 \geq B + C + U \geq B + (1 + 1/k)U.$$

The arithmetic-geometric mean inequality yields

$$\prod_{\text{active}} b_i \leq (U/k)^k \leq \left(\frac{1-B}{k+1} \right)^k.$$

The maximum of $B((1-B)/(k+1))^k$ on $[0, 1]$ occurs at $B = 1/(k+1)$ and equals M_k . Hence

$$B \prod_i \max(b_i, s) \leq \max_{0 \leq k \leq m-1} M_k s^{m-1-k}.$$

The relative projection deficit is therefore strictly larger than

$$\min_{1 \leq \ell \leq m} \frac{s^\ell}{M_{\ell-1}}. \quad (5.11)$$

This optimization is an upper bound on the denominator; it does not assert that every formal optimizer is compatible with every active coordinate regime.

Now suppose $y = (1-t)x + tc_P \notin K$, with $x \in P$ and $0 < t \leq 1$. Orient a strict separating affine functional so that K has larger values than y , subtract its minimum vertex value, and divide by the sum of its vertex values. This gives the preceding normalized b , with

$$b(y) = (1-t)b(x) + t/n \geq t/n.$$

Thus K lies strictly in $\{b > t/n\}$. Set $s = t/n$ in (5.11). If $t = \tau_d(\delta) > 0$, then every term $s^\ell/M_{\ell-1}$ is at least δ , a contradiction to (5.1). The argument includes $t = 1$. For $\delta = 0$, apply the positive- t argument and pass to the closed limit as before. $\square$

Proposition 5.7 (Uniform top-spectrum branch). *For every $d \geq 3$ and $0 \leq \rho \leq 1/3$, the term $\ell = m$ is largest in the definition of τ_d . Consequently*

$$\tau_d(\delta) = \gamma_d \rho, \quad \gamma_d = n \left(\frac{(d-2)^{d-2}}{(d-1)^{2d-3}} \right)^{1/(d-1)} = \frac{m+2}{m} \left(1 - \frac{1}{m} \right)^{(m-1)/m} \longrightarrow 1. \quad (5.12)$$

On this branch $\tau_d(\delta) \leq 2/3 < 1$, so the positive-core condition is automatic. In each fixed dimension the exact threshold

$$\delta \leq \min_{1 \leq \ell < m} \left(\frac{M_{m-1}^\ell}{M_{\ell-1}^m} \right)^{1/(m-\ell)} \quad (5.13)$$

is also sufficient and may enlarge this branch. In particular,

$$\tau_3(\delta) = \max(4\delta, \sqrt{2\delta}),$$

which equals $\sqrt{2}\rho$ for $\delta \leq 1/8$.

Proof. Set $f_\ell = \ell M_{\ell-1}^{1/\ell}$. Then $f_1 = 1$ and

$$f_\ell = (1 - 1/\ell)^{1-1/\ell} \quad (\ell \geq 2).$$

The latter sequence increases, since $x \log x$ increases on $[1/2, 1)$. In particular, $1/\sqrt{2} \leq f_m < 1$. For $\rho > 0$ and $2 \leq \ell \leq m$, the ratio of the ℓ th term to the m th term is

$$\frac{m}{\ell} \frac{f_\ell}{f_m} \rho^{m/\ell-1} \leq \frac{m}{\ell} \rho^{m/\ell-1} \leq 1 \quad (\rho \leq 1/3).$$

For the last inequality put $x = m/\ell \geq 1$ and use $\log x \leq x - 1 \leq (x - 1) \log 3$. For $\ell = 1$ the ratio is

$$\frac{m}{f_m} \rho^{m-1} \leq \sqrt{2} m 3^{1-m} \leq \frac{2\sqrt{2}}{3} < 1.$$

Indeed $m 3^{1-m}$ decreases for integer $m \geq 2$, because its successive ratio is $(m+1)/(3m) < 1$. The case $\rho = 0$ is immediate without division. Formula (5.12) follows by substitution, and

$$\tau_d(\delta) = \frac{m+2}{m} f_m \rho \leq \frac{2}{3} < 1.$$

For fixed m and $\ell < m$, the ratio $(\delta M_{\ell-1})^{1/\ell} / (\delta M_{m-1})^{1/m}$ tends to zero as δ decreases to zero. Rearranging the condition that this ratio be at most one for every $\ell < m$ gives (5.13). The $d = 3$ formula follows from $M_0 = 1$ and $M_1 = 1/8$. $\square$

5.4 Depth at faces of every positive dimension

Let $\lambda_0, \dots, \lambda_d$ be the barycentric coordinates of P . For a nonempty index set I define $f_I = \sum_{i \in I} \lambda_i$ and $s_I = \min_K f_I$.

Theorem 5.8 (Face-depth hierarchy). *If $|I| = r$ with $1 \leq r \leq d-1$, then*

$$F_{r,d-r}(s_I) \leq \delta, \quad F_{r,d-r}(s) = \sum_{j=r}^{d-1} \binom{d-1}{j} s^j (1-s)^{d-1-j}. \quad (5.14)$$

In particular,

$$s_I \leq \delta^{1/r} = \rho^{(d-1)/r}. \quad (5.15)$$

For each facet coordinate,

$$\min_K \lambda_i \leq 1 - (1 - \delta)^{1/(d-1)} \leq \frac{2\delta}{d-1} \quad \text{if } \delta \leq \frac{1}{2}. \quad (5.16)$$

Proof. The complement of I contains two indices a, b . Project along the edge direction $p_a - p_b$. The image D is an $(d-1)$ -simplex, obtained by identifying exactly the vertices a, b . The affine function f_I descends and becomes the sum of exactly r of the d barycentric coordinates of D . Since $K \subset \{f_I \geq s_I\}$, its projection omits $\{f_I < s_I\}$.

A direct section calculation in a simplex with d vertices gives the density of the sum of r barycentric coordinates as

$$\frac{(d-1)!}{(r-1)!(d-r-1)!} s^{r-1} (1-s)^{d-r-1}, \quad 0 < s < 1.$$

Its integral from zero to s is $F_{r,d-r}(s)$: differentiating the polynomial in (5.14) makes adjacent binomial terms telescope to the displayed density, and the polynomial vanishes at zero. No independence of barycentric coordinates is asserted. The omitted relative volume is therefore $F_{r,d-r}(s_I)$, proving the first inequality.

As a polynomial identity, $F_{r,d-r}(s)$ also equals the probability of at least r successes among $d-1$ independent Bernoulli trials with parameter s . The event that the first r trials succeed has probability s^r and is contained in that event. Thus $F_{r,d-r}(s) \geq s^r$, proving (5.15). For $r=1$, $F_{1,d-1}(s) = 1 - (1-s)^{d-1}$, which gives the exact first bound in (5.16). For $m = d-1$ and $\delta \leq 1/2$,

$$1 - (1-\delta)^{1/m} = \int_0^\delta \frac{1}{m} (1-u)^{1/m-1} du \leq \frac{\delta}{m(1-\delta)} \leq \frac{2\delta}{m}.$$

The zero and unit depth endpoints follow directly or by continuity. $\square$

The theorem concerns faces of positive dimension: the face $\{\lambda_i = 0 : i \in I\}$ has codimension $r \leq d-1$. The vertex case is separate, with projection exponent $d-1$, rather than d . When ρ is of order $1/d$, the facet estimate imposes exponentially small barycentric distance to each facet.

5.5 Necessary tests for finite hulls and the remaining target

Let K_0 be the convex hull of finitely many proposed near-corner points and candidate-simplex vertices. Since f_I is affine, $\min_{K_0} f_I$ is the minimum over the generators. Thus a violation of (5.14) rules out (5.1), regardless of any replacement minor constraints and without deciding whether the candidate simplex is maximum-volume.

Proposition 5.9 (A facet obstruction for finite hulls). *If every generator of K_0 has $\lambda_i \geq \varepsilon$ for some fixed i and $0 \leq \varepsilon \leq 1$, then any deficit δ satisfying (5.1) obeys*

$$\delta \geq 1 - (1-\varepsilon)^{d-1} \geq \frac{(d-1)\varepsilon}{1 + (d-1)\varepsilon}. \quad (5.17)$$

Proof. The first inequality is the facet case of Theorem 5.8. For $\varepsilon < 1$, $(1-\varepsilon)^{-(d-1)} \geq 1 + (d-1)\varepsilon$, which gives the second inequality after taking reciprocals. The endpoint $\varepsilon = 1$ follows directly. $\square$

For example, let the comparison simplex have barycentric matrix $Q_a = (1-a)I_n + (a/n)\mathbf{1}\mathbf{1}^\top$, and suppose every additional generator has every coordinate at least b/n . Then (5.17) applies to every facet with $\varepsilon = \min(a, b)/n$. If a and b are comparable to ρ , the actual deficit is bounded below by a constant times ρ , rather than ρ^{d-1} . Such thin positive-coordinate hulls cannot model the difficult $\rho \asymp 1/d$ regime.

The core theorem gives another finite test: whenever $\kappa_d \rho \leq 1$, every point

$$y_i = (1 - \kappa_d \rho) p_i + \kappa_d \rho c_P$$

must belong to K_0 . Membership is a finite linear feasibility problem. If membership fails, its separating functional determines, by Lemma 5.2, an actual projection direction violating the proposed deficit. This is only a necessary test; passing it does not establish all-direction projection control. The stronger core parameters give the corresponding stronger tests.

In particular, the full projection hypothesis permits the special central comparison matrix

$$Q_t = (1 - t)I_n + \frac{t}{n}\mathbf{1}\mathbf{1}^\top.$$

For a prescribed maximum simplex with barycentric matrix W , maximum volume still gives all replacement-minor inequalities for $W^{-1}Q_t$. A dimension-free lower bound on its entries, or a proof that the additional face-depth inequalities force such a bound, remains open here. These conclusions alone do not improve the general estimate $E(K, S) \leq 16n^2\rho$ to $Cn\rho$ and do not produce a genuine quadratic lower-bound body.

Corollary 5.10 (Concentric comparison without prescribed retention). *Let $t < 1$ be any core parameter proved above and put $T = (1 - t)P + tc_P$. Then T is a full-dimensional simplex with centroid c_P , and*

$$T \subset K \subset P = c_P + \frac{1}{1 - t}(T - c_P), \quad E(K, T) \leq \frac{t}{1 - t}. \quad (5.18)$$

For the sharper parameter $t = ((d + 1)/(d - 1))\rho \leq 1/2$, this gives $E(K, T) \leq 4\rho$. On the uniform finite-spectrum branch $\rho \leq 1/3$, it gives

$$E(K, T) \leq \frac{\gamma_d \rho}{1 - \gamma_d \rho}, \quad \gamma_d \rightarrow 1. \quad (5.19)$$

Proof. The core inclusion and $K \subset P$ give the sandwich, and the displayed homothety is exact. The defining centered excess therefore gives (5.18). If $t \leq 1/2$, then $t/(1 - t) \leq 2t \leq 4\rho$. The finite-spectrum assertion follows by substituting $t = \gamma_d \rho$ and using Proposition 5.7. $\square$

Remark 5.11 (Actual-defect substitution and scope). On the small-defect branch where the previously established actual enclosing-simplex/projection chain supplies $\delta = H_d e(K)$ with $H_d = 3d^2(d + 1)^3(d + 2)$, one may substitute

$$t = \frac{d + 1}{d - 1} H_d^{1/(d-1)} e(K)^{1/(d-1)}$$

in (5.18) whenever $t < 1$. Likewise the finite-spectrum parameter may be applied to the same actual projection packet, within its stated gate. Neither substitution changes the defect or its power law. The simplex T retains the enclosing centroid c_P ; it is not identified with an arbitrarily prescribed maximum simplex S , and its centroid is not substituted for the original centroid of S . Thus the dimension-free comparison approximation and prescribed-maximum retention remain distinct assertions.

6 Inverse minor rigidity in structured classes

Let W be an invertible nonnegative column-stochastic $n \times n$ matrix, and write

$$B = W^{-1}, \quad M_k(B) = \max_{|I|=|J|=k} |\det B_{I,J}|, \quad a(B) = \max\{0, -\min_{i,j} B_{ij}\}.$$

The general problem asks for a dimension-free bound $a(B) \leq C(c - 1)$ when $M_k(B) \leq c^k$ for every $1 \leq k \leq n$, with $c \geq 1$. We prove such bounds for support-two matrices, low-order blocks, and symmetric positive-definite matrices. The first class yields an all-scale geometric estimate for every prescribed maximum simplex of an edge-supported body. The unrestricted inverse-minor problem is not resolved by these results.

6.1 Weighted cycles

Lemma 6.1 (Weighted cyclic blocks). *Suppose a column-stochastic block of order $q \geq 2$ has the form*

$$W = (I_q + PR_d)D, \quad D = \text{diag}(d_i), \quad R_d = \text{diag}(r_i), \quad d_i = \frac{1}{1+r_i}, \quad r_i > 0,$$

where P is a cyclic permutation matrix. If $M_1(B) \leq c$ and $M_2(B) \leq c^2$, then

$$a(B) \leq c^2 - 1 \quad (1 \leq c < \sqrt{2}), \quad (6.1)$$

$$a(B) \leq (2 + \sqrt{2})(c - 1) \quad (c \geq 1). \quad (6.2)$$

Proof. Let $R = \prod_i r_i$. Interchanging the two cyclic matchings by a row permutation replaces this product by R^{-1} . We may choose the orientation with $R \leq 1$. In even order, $R = 1$ is excluded by invertibility. Thus

$$L = 1 - (-1)^q R > 0.$$

The finite geometric identity gives

$$B = \frac{1}{L} D^{-1} [I_q - PR_d + (PR_d)^2 - \cdots + (-1)^{q-1} (PR_d)^{q-1}]. \quad (6.3)$$

In particular,

$$B_{ii} = \frac{1+r_i}{L}, \quad \det B_{\{i,j\},\{i,j\}} = \frac{(1+r_i)(1+r_j)}{L} \quad (i \neq j). \quad (6.4)$$

For the second formula, Jacobi's complementary-minor identity applies. Directly, when $q > 2$, deleting vertices i, j from the directed cycle leaves an acyclic principal submatrix, triangular after topological ordering, with determinant the product of its d_ℓ . For $q = 2$, the formula is simply $\det B = 1/(d_1 d_2 L)$.

First suppose every $r_i \leq 1$. If q is odd, then $R \leq r_i$, so $B_{ii} \geq 1$; if q is even, $L \leq 1$ gives the same conclusion. In row i , each absolute inverse entry in (6.3) is the common factor $(1+r_i)/L$ times a product of consecutive ratios. Negative entries correspond to odd-length paths. Every such product is at most the last one-edge factor r_j , where j is the predecessor of i . Thus the largest negative magnitude in that row occurs on the incoming edge and equals

$$-B_{ij} = \frac{(1+r_i)r_j}{L}.$$

The principal two-minor on i, j is

$$\frac{(1+r_i)(1+r_j)}{L} = B_{ii} + (-B_{ij}) \geq 1 + (-B_{ij}).$$

It follows that $a(B) \leq M_2(B) - 1 \leq c^2 - 1$.

Next suppose some ratio is at least one, and let $x \geq y$ be the two largest ratios, with $x \geq 1$. Then $R \leq xy$: if $y \leq 1$ all the remaining ratios are at most one, while if $y \geq 1$ then $xy \geq 1 \geq R$. In odd order,

$$\begin{aligned} (1+x)(1+y) - 2(1+R) &\geq (1+x)(1+R/x) - 2(1+R) \\ &= (x-1)(1-R/x) \geq 0. \end{aligned}$$

The corresponding two-minor in (6.4) is therefore at least two. In even order $L \leq 1$ and $(1+x)(1+y) \geq 2$ give the same result. Hence this case requires $c \geq \sqrt{2}$.

Consequently $c < \sqrt{2}$ forces the first case and proves (6.1). For $c \geq \sqrt{2}$, the entry bound $a(B) \leq c$ yields

$$\frac{a(B)}{c-1} \leq \frac{c}{c-1} \leq 2 + \sqrt{2}.$$

For $1 < c < \sqrt{2}$, the first case gives $a(B)/(c-1) \leq c+1 < 1 + \sqrt{2}$. At $c = 1$ its conclusion is $a(B) = 0$. This proves (6.2). $\square$

Independent row and column permutations of W merely permute the rows and columns of B independently and change signs of its minors. All these inequalities are therefore invariant under such permutations. Zero edge weights may be included by continuity, but the support decomposition below handles them without treating a zero-weight edge as part of a genuine cycle.

6.2 Columns supported on at most two rows

Theorem 6.2 (Support-two inverse-minor rigidity). *Suppose every column of W has at most two nonzero entries. If $M_1(B) \leq c$ and $M_2(B) \leq c^2$, then*

$$a(B) \leq c^2 - 1 \quad (1 \leq c < \sqrt{2}), \quad (6.5)$$

$$a(B) \leq (2 + \sqrt{2})(c - 1) \quad (c \geq 1). \quad (6.6)$$

For $n = 1$ only the order-one bound is needed. The constants do not depend on the matrix order, cycle lengths, or sizes of attached trees.

Proof. Associate a multigraph to W . Its vertices are the row indices; a column with two positive entries is an edge between its two supporting rows, and a singleton column is a loop at its supporting row. Parallel edges are allowed. Connected components partition the rows and columns into matrix blocks. Each block must be square: if a component has unequal row and column counts, summing the component rank bounds, each at most the smaller count, makes the whole square matrix singular. Every connected component therefore has equally many edges and vertices and is unicyclic, including loops and doubled edges as cycles of lengths one and two.

Remove the trees attached to the unique cycle. Pair each non-core vertex v with the edge joining it to its parent $p(v)$ towards the core. Order core rows and columns first and tree vertices in parent-before-child order. Then

$$W = \begin{pmatrix} W_C & F \\ 0 & T \end{pmatrix},$$

where T is triangular. The column paired with v has weight $d_v > 0$ at v and $1 - d_v$ at its parent. Its full inverse diagonal entry is $1/d_v$, so

$$r_v := \frac{1 - d_v}{d_v} = \frac{1}{d_v} - 1 \leq c - 1. \quad (6.7)$$

The core inverse $B_C = W_C^{-1}$ is a submatrix of B and inherits the order-one and order-two bounds. A loop core has $B_C = [1]$; the other cores satisfy Lemma 6.1.

For completeness, let $b_v = Be_v$ denote an inverse column. The identity $BWe_v = e_v$ gives the exact recursion

$$b_v = d_v^{-1}e_v - r_v b_{p(v)}. \quad (6.8)$$

For a core index p , b_p has core part $B_C e_p$ and zero tree part. Repeatedly applying (6.8) shows that every off-diagonal entry in a tree column is a product of at least one tree ratio, times a

core inverse entry or an ancestor diagonal entry $1/d_u$. Every such starting factor has absolute value at most c . Equivalently, the tree block has the form $(I + N)D$ with $N_{p(v),v} = r_v$, and the finite nilpotent expansion of $(I + N)^{-1}$ gives these unique path products.

If $c < \sqrt{2}$, then $0 \leq r_v \leq c - 1 < 1$. Every tree-column off-diagonal magnitude is therefore at most $c(c - 1) \leq c^2 - 1$. Tree diagonal entries are positive, and core negative entries satisfy the same $c^2 - 1$ bound by the cyclic lemma. This proves (6.5). For $c \geq \sqrt{2}$, the bound $a(B) \leq c$ gives (6.6) as in that lemma; for smaller c the local bound already implies it. Taking direct sums and undoing the permutations completes the proof. $\square$

6.3 Retention of a prescribed maximum simplex

The next reduction records why the matrix estimates control the entire body, including points beyond the selected near-vertex comparison simplex.

Lemma 6.3 (From near-vertex points to inverse minors). *Let P have n vertices, let $K \subset P$ be a convex body, and let S be a prescribed maximum-volume simplex in K . Let W be the matrix of its vertex coordinates in the barycentric system of P . Suppose K contains n points whose nonnegative column-stochastic barycentric matrix Q satisfies $Q_{ii} \geq 1 - \rho$. For $0 \leq \rho < 1/2$, put*

$$B = W^{-1}, \quad A = BQ, \quad c = (1 - 2\rho)^{-1}.$$

Then every square minor of A has absolute value at most one, and $M_k(B) \leq c^k$ for every k . Moreover,

$$E(K, S) \leq n \min\{1, a(B)\}. \quad (6.9)$$

The latter conclusion applies to all of K and uses the original centroid of the prescribed S .

Proof. Replacing any selected set of vertices of S by selected columns of Q gives a simplex in K . The determinant ratio is the corresponding square minor of A , up to sign. Maximality of S therefore gives the all-minor bound. Since $\|Q - I_n\|_1 \leq 2\rho$, the Neumann series gives $\|Q^{-1}\|_1 \leq c$. From $B = AQ^{-1}$, Cauchy–Binet bounds each k -minor of B by the sum of absolute k -minors of Q^{-1} with a fixed column set. Expanding their determinants over injections bounds that sum by the product of the corresponding k absolute column sums, at most c^k .

For any $x \in K$, its P -barycentric vector z is nonnegative and sums to one. Its S -barycentric vector is Bz , whose every entry is at least $\min B$. Single-vertex replacement maximality also gives $|\alpha_i^S(x)| \leq 1$. The original-centroid dilation criterion (B.8) is $\alpha_i^S(x) \geq -E/n$. Combining these two lower bounds for every $x \in K$ proves (6.9). $\square$

Theorem 6.4 (Edge-supported prescribed simplex). *Under the hypotheses of Lemma 6.3, suppose every vertex of the originally prescribed maximum simplex S lies on an edge of P , allowing vertices of P themselves. For every $\rho \geq 0$ for which the stated near-vertex hypothesis holds,*

$$E(K, S) \leq (4 + 2\sqrt{2})n\rho. \quad (6.10)$$

No bound on the number of vertices of K is required.

Proof. The support assumption says every column of W has at most two nonzero entries. Put

$$\rho_0 = \frac{1 - 1/\sqrt{2}}{2}.$$

For $0 \leq \rho < \rho_0$, the reduction has $c = (1 - 2\rho)^{-1} < \sqrt{2}$. By Theorem 6.2 and (6.9),

$$\frac{E(K, S)}{n} \leq c^2 - 1 = \frac{4\rho(1 - \rho)}{(1 - 2\rho)^2} \leq (4 + 2\sqrt{2})\rho.$$

Indeed $4(1 - \rho)/(1 - 2\rho)^2$ increases on this interval and equals $4 + 2\sqrt{2}$ at ρ_0 . For $\rho \geq \rho_0$, universal single-replacement maximality gives $E(K, S) \leq n$, and $1/\rho_0 = 4 + 2\sqrt{2}$. This proves the entire large- ρ branch, including $\rho \geq 1/2$ without using the Neumann reduction. $\square$

Theorem 6.5 (Every prescribed maximizer of an edge-supported body). *Let $K \subset P$ be a full-dimensional polytope whose every vertex lies on an edge of P . Suppose K contains the near-vertex points in Lemma 6.3. Then (6.10) holds for every prescribed maximum-volume simplex S of K , even if its own vertices are not edge-supported.*

Proof. Let $v_1, \dots, v_N$ be the vertices of K . Write every vertex s_j of the prescribed S as a convex combination $s_j = \sum_{\ell} \lambda_{\ell j} v_{\ell}$. Independently sample $T_j = v_{\ell}$ with probabilities $\lambda_{\ell j}$. Use homogeneous affine coordinates, in which the determinant of the n columns is proportional to oriented simplex volume, and order S so that $\det S = D > 0$. Determinant multilinearity gives

$$\mathbb{E}[\det T] = \det S = D.$$

Maximality gives $|\det T| \leq D$ for every sampled tuple. Hence every positive-probability tuple has $\det T = D$: it is nondegenerate, has the same orientation, and is a maximum vertex simplex.

Fix $x \in K$ and an index j . Cramer's rule and multilinearity in all unreplaced columns give

$$\begin{aligned} \alpha_j^S(x) &= \frac{\det(S \text{ with column } j \text{ replaced by } x)}{D} \\ &= \frac{\mathbb{E}[\det(T \text{ with column } j \text{ replaced by } x)]}{D} = \mathbb{E}[\alpha_j^T(x)]. \end{aligned} \quad (6.11)$$

Each sampled T is edge-supported. The coordinate bound in the proof of Theorem 6.4 therefore gives $\alpha_j^T(x) \geq -(4 + 2\sqrt{2})\rho$ for all x, j . Averaging proves the identical coordinate bound for the original S . The dilation criterion then yields (6.10) with S and its own centroid. No sampled centroid is substituted. $\square$

Corollary 6.6 (Compact bodies supported on the one-skeleton). *Theorem 6.5 also applies when K is a compact convex body and every extreme point of K lies on the finite one-skeleton of P .*

Proof. For every edge F of P , $K \cap F$ is empty, a point, or a closed interval. A point in the relative interior of a nontrivial such interval is not extreme in K . Thus each edge contains at most two extreme points of K , its interval endpoints. There are only finitely many edges, so K has finitely many extreme points and, by the finite-dimensional compact convex hull theorem, is their convex hull. It is therefore a polytope to which Theorem 6.5 applies. In particular, the same conclusion holds when K is the convex hull of finitely many closed segments on the edges of P . $\square$

6.4 A conditional corollary for the actual defect

Corollary 6.7 (Actual-defect retention with compatible edge support). *Retain the original actual invariant*

$$e(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\text{pyr } K)}{R(K)} - 1 - \frac{1}{d+1}, \quad R(L) = \frac{\text{Vol}(\Pi L)}{\text{Vol}(L)^{\dim L - 1}}, \quad (6.12)$$

and put $H_d = 3d^2(d+1)^3(d+2)$, $n = d+1$, $d \geq 3$. On the branch $0 \leq e(K) \leq H_d^{-1}$, assume additionally that at least one enclosing simplex P admitted by the actual-defect, projection,

and cap chain has every extreme point of K on its one-skeleton. Then for every originally prescribed maximum simplex S ,

$$E(K, S) \leq (4 + 2\sqrt{2})(d + 1)H_d^{1/(d-1)}e(K)^{1/(d-1)}. \quad (6.13)$$

In particular,

$$E(K, S) \leq (768 + 384\sqrt{2})de(K)^{1/(d-1)}. \quad (6.14)$$

For $e(K) > H_d^{-1}$, both bounds hold without the compatibility or edge-support hypothesis.

Proof. On the stated small-defect branch, the retained actual chain supplies the compatible enclosing P , relative projection deficit at most $\delta = H_de(K)$, and near-vertex radius $\rho = \delta^{1/(d-1)}$. The extreme-point assumption makes K a polytope by Corollary 6.6. Substitution into Theorem 6.5 gives (6.13); the zero-defect endpoint is included by the zero-radius cap conclusion. The retained estimate $H_d^{1/(d-1)} \leq 144$, together with $d + 1 \leq 4d/3$, gives (6.14), since $192(4 + 2\sqrt{2}) = 768 + 384\sqrt{2}$. For completeness, the numerical bound on H_d follows from $H_3 = 8640 < 144^2$ and $H_{d+1}/H_d \leq 25/6 < 144$ for $d \geq 3$, by induction. If $e(K) > H_d^{-1}$, then $H_d^{1/(d-1)}e(K)^{1/(d-1)} > 1$, so the universal $E(K, S) \leq d + 1$ proves (6.13) directly; the coarser bound follows. $\square$

Remark 6.8 (The compatibility quantifier). The hypothesis refers to a P produced by the actual-defect chain. The existence of some unrelated enclosing simplex with edge-supported extreme points is insufficient. No result here shows that the chain's P can always be chosen with this property. If only one prescribed S is to be retained, it is enough that this S be edge-supported relative to an admissible P , by Theorem 6.4. The body-level hypothesis is what gives the conclusion simultaneously for every prescribed maximum simplex without inspecting each support. The exact coefficient in (6.13) is asymptotic to $(4 + 2\sqrt{2})d$, since $H_d^{1/(d-1)} \rightarrow 1$. These are conditional corollaries of the original invariant and the retained upstream chain, not an unconditional theorem for all convex bodies.

6.5 Order three and an exact golden-ratio obstruction

Theorem 6.9 (An order-three two-minor inequality). *If $n = 3$, then*

$$1 - \min_{i,j} B_{ij} \leq M_2(B). \quad (6.15)$$

Consequently, if $M_1(B) \leq c$ and $M_2(B) \leq c^2$, then

$$a(B) \leq \min\{c, c^2 - 1\} \leq \varphi^2(c - 1), \quad \varphi = \frac{1 + \sqrt{5}}{2}. \quad (6.16)$$

The coefficient φ^2 is optimal. The same conclusions hold for direct sums of blocks of orders at most three, after independent row and column permutations.

Proof. Permutations preserve stochasticity and the quantities in (6.15). First arrange $D = \det W > 0$. Independent even row and column permutations are transitive on the three indices, so any chosen entry of B can be placed at position $(3, 1)$ while preserving $D > 0$. Write

$$W = \begin{pmatrix} u & v & z \\ r & s & t \\ 1 - u - r & 1 - v - s & 1 - z - t \end{pmatrix}.$$

If C_{13} is the cofactor at row 1, column 3, direct expansion gives

$$\begin{aligned} D - C_{13} &= (s - r)(1 - z) + t(v - u) \\ &= (1 - z - t)(s - r) + t[(v + s) - (u + r)] + z \cdot 0. \end{aligned}$$

This is a convex combination, with the third-column weights $1 - z - t, t, z$, of the values $s - r$, $W_{31} - W_{32}$, and 0. Each is at most $w_* := \max_{i,j} W_{ij}$. Since $B_{31} = C_{13}/D$,

$$1 - B_{31} \leq \frac{w_*}{D}.$$

Jacobi's complementary-minor identity gives $M_2(B) = w_*/D$. Applying the argument to every entry proves (6.15). Thus $a(B) \leq c^2 - 1$, and the entry bound gives $a(B) \leq c$. For $c > 1$, the upper envelope of $\min\{c, c^2 - 1\}/(c - 1)$ has maximum φ^2 at the crossing $c = \varphi$; $c = 1$ follows directly.

A one-by-one block is [1]. In order two, every column-stochastic inverse can be written

$$\begin{pmatrix} x & y \\ 1 - x & 1 - y \end{pmatrix}.$$

Each negative entry has a positive entry in its column larger in absolute value by one. Hence $a(B) \leq M_1(B) - 1 \leq c - 1$ in that order. Applying the appropriate bound in each block proves the direct-sum assertion. Optimality follows from the next exact example. $\square$

Proposition 6.10 (An all-order golden-ratio extremizer). *Let $p = \varphi^{-1}$ and set*

$$W = \begin{pmatrix} 0 & p & 1/2 \\ p & 0 & 1/2 \\ 1 - p & 1 - p & 0 \end{pmatrix}.$$

Then W is nonnegative and column-stochastic, $\det W = p(1 - p) = \varphi^{-3}$, and

$$B = W^{-1} = \begin{pmatrix} -\varphi/2 & \varphi/2 & \varphi^2/2 \\ \varphi/2 & -\varphi/2 & \varphi^2/2 \\ 1 & 1 & -\varphi \end{pmatrix}.$$

Its all-order root bound is exactly $c = \varphi$, and

$$M_1(B) = \varphi, \quad M_2(B) = \varphi^2, \quad M_3(B) = \varphi^3, \quad a(B) = \varphi = c^2 - 1 = \varphi^2(c - 1). \quad (6.17)$$

Thus any universal estimate $a(B) \leq C(c - 1)$ valid for all $c \geq 1$ must have $C \geq \varphi^2$.

Proof. The stochasticity and displayed inverse follow by direct multiplication using $\varphi^2 = \varphi + 1$. The largest absolute entry of B is φ . Jacobi gives $M_2(B) = p/\det W = \varphi^2$, and $M_3(B) = 1/\det W = \varphi^3$. The minimum entry is $-\varphi$, which yields all equalities in (6.17). $\square$

This is an exact obstruction to a smaller global constant. It is not a counterexample in the limit $c \downarrow 1$. The order-three inequality (6.15) is dimension-specific; the corresponding proposed estimate with $\max\{M_1(B), M_2(B)\}$ is false in order five. In particular, an unrestricted extension based only on first- and second-order minors would require a new argument.

6.6 Positive definiteness and a local permutation basin

Theorem 6.11 (Symmetric positive-definite case). *If W is symmetric and positive definite, then $M_1(B) \leq c$ alone implies*

$$a(B) \leq c - 1. \quad (6.18)$$

The coefficient one is sharp even under all-order minor bounds. In the setting of Lemma 6.3, a symmetric positive-definite prescribed W gives the all-scale bound

$$E(K, S) \leq 4n\rho. \quad (6.19)$$

Proof. A nonnegative column-stochastic matrix has spectral radius at most one. Symmetry and positive definiteness put all eigenvalues of W in $(0, 1]$. Therefore $H = B - I_n$ is positive semidefinite and

$$0 \leq H_{ii} = B_{ii} - 1 \leq c - 1.$$

Every two-coordinate positive-semidefinite restriction gives, for $i \neq j$,

$$|B_{ij}| = |H_{ij}| \leq \sqrt{H_{ii}H_{jj}} \leq c - 1.$$

The diagonal entries are at least one, so this proves (6.18).

For sharpness take $W = \begin{pmatrix} 1-s & s \\ s & 1-s \end{pmatrix}$ with $0 < s < 1/2$. Its inverse has negative magnitude $a = s/(1-2s)$, diagonal $1+a$, and determinant $1+2a$. Since $(1+a)^2 \geq 1+2a$, its all-order root bound is $c = 1+a$. Thus equality holds in (6.18).

For the geometric claim, if $0 \leq \rho < 1/2$ then (6.9) gives $E(K, S)/n \leq \min\{1, c-1\}$ with $c = (1-2\rho)^{-1}$. For $\rho \leq 1/4$, $c-1 = 2\rho/(1-2\rho) \leq 4\rho$; for $\rho \geq 1/4$ the universal bound $E(K, S)/n \leq 1 \leq 4\rho$ applies, also beyond the Neumann range. $\square$

The symmetry and positive-definiteness assumption is coordinate-dependent. It must be verified and does not follow from positivity of the entries of W .

Lemma 6.12 (A local permutation basin). *Suppose independent row and column permutations put W in a form with*

$$s = \max_j (1 - W_{jj}) < \frac{1}{4}.$$

Then $M_1(B) \leq c$ implies

$$a(B) \leq \frac{c-1}{1-4s}. \quad (6.20)$$

In particular, $s \leq 1/8$ gives $a(B) \leq 2(c-1)$.

Proof. Put $H = I_n - W$. Its induced column norm is $\|H\|_1 = 2s$, and its maximum entry modulus is at most s . For $k \geq 1$,

$$\max_{i,j} |(H^k)_{ij}| \leq s(2s)^{k-1}.$$

The Neumann series therefore gives $a(B) \leq s/(1-2s)$. At an index j with $H_{jj} = s$, it also gives

$$B_{jj} \geq 1 + s - \frac{2s^2}{1-2s} = 1 + \frac{s(1-4s)}{1-2s}.$$

Since $B_{jj} \leq c$, combining these inequalities proves (6.20). The case $s = 0$ is immediate. $\square$

This lemma does not assert the existence of a suitable permutation. A dimension-free argument placing arbitrary instances in such a basin would itself constitute a substantive missing step.

6.7 Additional all-order reductions

The following results preserve useful information beyond the support-two theorem. Here the hypothesis is again $M_k(B) \leq c^k$ for every relevant order.

Proposition 6.13 (Localized minor completion and occupancy). *Let $R, S \subset [n]$ be disjoint row-label sets, with $|R| = k$ and $|S| = \ell$, and let $J \subset [n]$ have size k . For each $s \in S$, independently draw Y_s from the distribution $W_{:,s}$. Then*

$$|\det B_{R,J}| \leq c^{k+\ell} \mathbb{P}(Y_s \text{ are pairwise distinct and all outside } J). \quad (6.21)$$

In particular, for any B_{ij} and $S \subset [n] \setminus \{i\}$,

$$\begin{aligned} |B_{ij}| &\leq c^{|S|+1} \mathbb{P}(Y_s \text{ are distinct and all different from } j) \\ &\leq c^{|S|+1} \prod_{s \in S} (1 - W_{j,s}). \end{aligned} \quad (6.22)$$

With $k = 0$ and the empty determinant equal to one,

$$1 \leq c^{|S|} \mathbb{P}(Y_s \text{ are pairwise distinct}). \quad (6.23)$$

Proof. Order $R \cup S$ with R first. Form the $(k + \ell)$ -square matrix whose first k columns are $B_{R \cup S, J}$ and whose last ℓ columns are the coordinate vectors e_s restricted to $R \cup S$. Its determinant is $\det B_{R,J}$ up to a fixed ordering sign. Substitute

$$e_s = \sum_h W_{h,s} B_{:,h}$$

and expand multilinearly in the last ℓ columns. A term vanishes if its selected indices repeat or if one belongs to J . Every remaining determinant is a $(k + \ell)$ -minor of B , bounded in magnitude by $c^{k+\ell}$. The sum of the corresponding nonnegative product weights is exactly the probability in (6.21). Taking $k = 1$ gives the first entry bound. Dropping the collision-free condition and using the independent draws gives the product bound. Taking $k = 0$ gives (6.23). $\square$

The collision-free term retains both the fixed minor and its forbidden column set and can be stronger than the displayed product bound.

Proposition 6.14 (Regularization of a uniformly almost nonnegative inverse). *Suppose A is invertible, every square minor of A has absolute value at most one, $\mathbf{1}^\top A = \mathbf{1}^\top$, and $V = A^{-1}$ satisfies $V_{ij} \geq -\eta/n$ for some $\eta \geq 0$. With $J = \mathbf{1}\mathbf{1}^\top$, define*

$$W = \frac{V + (\eta/n)J}{1 + \eta}, \quad Q = (1 + \eta)I_n - \frac{\eta}{n}J, \quad B = W^{-1} = AQ.$$

Then W is nonnegative and column-stochastic,

$$|B_{ij} - A_{ij}| \leq 2\eta, \quad M_k(B) \leq (1 + 2\eta)^k \quad (1 \leq k \leq n). \quad (6.24)$$

Thus a universal bound $a(B) \leq C(c - 1)$ in the stochastic inverse-minor problem would imply $-\min A \leq (2C + 2)\eta$ under these hypotheses.

Proof. The assumed lower bound gives $W \geq 0$, and $\mathbf{1}^\top V = \mathbf{1}^\top$ gives its column sums. Using the same column-sum identity for A verifies $B = AQ$. Entrywise,

$$B_{ij} - A_{ij} = \eta \left(A_{ij} - \frac{1}{n} \sum_h A_{ih} \right).$$

The order-one minor bound on A proves the first part of (6.24).

For a fixed k -element column set, the only nonzero k -minors of Q have the same row set or exchange one index. The principal minor is

$$(1 + \eta)^{k-1}(1 + \eta - k\eta/n),$$

and each of the $k(n - k)$ one-index-exchange minors has magnitude $(\eta/n)(1 + \eta)^{k-1}$. Cauchy–Binet therefore gives

$$\begin{aligned} |\det B_{I,J}| &\leq (1 + \eta)^{k-1} \left[1 + \eta + \frac{\eta k(n - k - 1)}{n} \right] \\ &\leq (1 + \eta)^{k-1} [1 + (k + 1)\eta] \leq (1 + 2\eta)^k. \end{aligned}$$

For the last inequality, Bernoulli's inequality gives

$$\frac{(1 + 2\eta)^k}{(1 + \eta)^{k-1}} = (1 + \eta) \left(1 + \frac{\eta}{1 + \eta} \right)^k \geq 1 + (k + 1)\eta.$$

Applying the hypothetical stochastic bound with $c = 1 + 2\eta$ and then the entrywise difference estimate gives $-\min A \leq 2C\eta + 2\eta$, as asserted. $\square$

This is a one-way reduction. It does not establish that an arbitrary stochastic instance admits the stronger regularized representation in the opposite direction.

Lemma 6.15 (Conditional stability from a dominant matching). *If, after a row permutation, $W_{ii} \geq 1 - \varepsilon$ for every i , with $0 \leq \varepsilon < 1/2$, then*

$$\|W^{-1} - I_n\|_1 \leq \frac{2\varepsilon}{1 - 2\varepsilon}. \quad (6.25)$$

In particular the same upper bound holds for the negative entry magnitudes in the permuted inverse.

Proof. Nonnegativity and column stochasticity give $\|I_n - W\|_1 \leq 2\varepsilon$. The Neumann series yields

$$\|W^{-1} - I_n\|_1 \leq \frac{\|I_n - W\|_1}{1 - \|I_n - W\|_1} \leq \frac{2\varepsilon}{1 - 2\varepsilon}.$$

Every negative entry is bounded by this norm, proving the claim. $\square$

A dimension-free dominant-matching theorem of the form $W_{i,\sigma(i)} \geq 1 - O(c - 1)$ would settle the target through this lemma. No such matching theorem is established here.

6.8 Limitations of the structural results

The support-two proof uses the unicyclic graph decomposition essentially. A column supported on three or more rows does not reduce to the same path recursion, and no reduction of arbitrary W to this class is supplied. The all-order conjecture $a(B) \leq c^2 - 1$ remains open in general; its analogue based only on first- and second-order bounds is false in order five. Neither an unrestricted dimension-free inverse-minor theorem nor a counterexample to it follows from this section. The special central- Q_t formulation from Section 5 and the unrestricted prescribed-maximum linear dimension bound also remain open. In particular, a matrix construction by itself is not an actual-defect lower-bound body: the all-direction projection and enclosing-simplex requirements must still be verified.

A A finite determinant assignment theorem

We record a finite version of the retained volume-weighted assignment argument, with its proof. For a real number t , write $t_+ = \max(t, 0)$. All probability measures in this section have finite support, so no integrability or limiting-law issue is present.

Lemma A.1 (Seminorm assignment). *Let μ be a centered probability measure with finite support $\mathcal{A} \subset \mathbb{R}^d$ whose affine span is $\mathbb{R}^d$. Let N be a seminorm with $N(x) \leq R_0$ on $\mathcal{A}$, where $R_0 \geq 0$. For independent samples with law μ , put*

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}|\det((X_0, 1), \dots, (X_d, 1))|.$$

Then $B > 0$, $D := B - A \geq 0$, and there are affinely independent anchors $w_0, \dots, w_d \in \mathcal{A}$ and an assignment $I : \mathcal{A} \rightarrow \{0, \dots, d\}$ with

$$h := \mathbb{E}N(X - w_{I(X)}) \leq R_0(d+1)(d+2)\frac{D}{B}. \quad (\text{A.1})$$

One can choose $I(x)$ to be the least index of a largest barycentric coordinate of x in the anchor simplex.

Proof. For an ordered base $\mathbf{x} = (x_1, \dots, x_d)$ define

$$F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1)), \quad P_{\mathbf{x}} = \mathbb{E}(F_{\mathbf{x}}(X))_+, \quad M_{\mathbf{x}} = \mathbb{E}(-F_{\mathbf{x}}(X))_+.$$

Centering and affinity in the last column give $|\mathbb{E}F_{\mathbf{x}}(X)| = |\det(x_1, \dots, x_d)|$. Hence

$$D = 2\mathbb{E}_{\mathbf{X}} \min(P_{\mathbf{X}}, M_{\mathbf{X}}) \geq 0. \quad (\text{A.2})$$

For two test points set

$$\psi_{\mathbf{x}}(y, z) = \min((F_{\mathbf{x}}(y))_+, (-F_{\mathbf{x}}(z))_+) + \min((-F_{\mathbf{x}}(y))_+, (F_{\mathbf{x}}(z))_+).$$

Each conditional expected minimum is at most $\min(P_{\mathbf{x}}, M_{\mathbf{x}})$. Therefore

$$\mathbb{E}_{\mathbf{X}, Y, Z} \psi_{\mathbf{X}}(Y, Z) \leq D. \quad (\text{A.3})$$

These statements include singular bases.

For an anchor tuple $W = (w_0, \dots, w_d)$, put $v(W) = |\det((w_0, 1), \dots, (w_d, 1))|$. Form $G(W, x)$ by adding the following witnesses, with any fixed order on each base:

- (i) for each i , the base is the other d anchors and the test points are w_i, x ;
- (ii) for each $i < j$, the base is x and the other $d - 1$ anchors, and the test points are w_i, w_j .

Reversing base orientation does not change ψ . There are $M_d = (d+1)(d+2)/2$ summands. Each uses $d+2$ independent samples when W and x are sampled, so, with $J(W) = \mathbb{E}_X G(W, X)$,

$$\mathbb{E}_W J(W) \leq M_d D, \quad \mathbb{E}_W v(W) = B > 0.$$

Positivity of B follows from an affinely independent tuple in the support. Weighting anchor tuples by $v(W)/B$ gives an anchor tuple with $v(W) > 0$ and

$$\frac{J(W)}{v(W)} \leq M_d \frac{D}{B}. \quad (\text{A.4})$$

Indeed its weighted mean is $\mathbb{E}[J(W)\mathbf{1}_{\{v(W)>0\}}]/B \leq M_d D/B$. No assertion that J vanishes at singular tuples is needed.

Fix this tuple, and write $\alpha_i(x)$ for its barycentric coordinates. Determinant affinity gives, for witnesses of type (i), $v(W) \min(1, (-\alpha_i)_+)$; those of type (ii) equal

$$v(W)[\min((\alpha_i)_+, (\alpha_j)_+) + \min((-\alpha_i)_+, (-\alpha_j)_+)].$$

It follows that $G(W, x) \geq v(W)\phi(x)$, where

$$\phi(x) = \sum_i \min(1, (-\alpha_i(x))_+) + \sum_{i < j} \min((\alpha_i(x))_+, (\alpha_j(x))_+).$$

Let $j = I(x)$ be a largest coordinate. Since $\sum_i \alpha_i = 1$, we have $\alpha_j > 0$. Put $N_x = \sum_i (-\alpha_i)_+$ and $U_x = \sum_{i \neq j} (\alpha_i)_+$. The support bounds yield both

$$N(x - w_j) \leq 2R_0(N_x + U_x), \quad N(x - w_j) \leq 2R_0.$$

The first follows by expanding $x - w_j = \sum_{i \neq j} \alpha_i(w_i - w_j)$; the second uses $x, w_j \in \mathcal{A}$. The pairs containing the largest coordinate give $U_x \leq \sum_{i < l} \min((\alpha_i)_+, (\alpha_l)_+)$. Also $\min(1, N_x) \leq \sum_i \min(1, (-\alpha_i)_+)$. Thus

$$N(x - w_{I(x)}) \leq 2R_0 \min(1, N_x + U_x) \leq 2R_0 \phi(x) \leq 2R_0 \frac{G(W, x)}{v(W)}.$$

Integrating and using (A.4) proves (A.1). The reasoning includes $D = 0$ without division by D . $\square$

B From the actual facet law to an enclosing simplex

B.1 Raw facet identities and the original defect

For a full-dimensional p -polytope L , let $u_i = s_i n_i$ be its facet area-normal vectors and $b_i = s_i h_L(n_i)$, where n_i is the outward unit normal and $s_i > 0$ is the facet area. Set

$$\mathcal{H}(L) = \sum_{|J|=p} |\det(u_j : j \in J)|, \quad \mathcal{L}(L) = \sum_{|J|=p+1} |\det((u_j, b_j) : j \in J)|. \quad (\text{B.1})$$

The order chosen within a subset is immaterial. Translating L replaces b_i by $b_i - \langle u_i, z \rangle$ and leaves these sums unchanged.

Cauchy's formula gives $b_L(z) = \frac{1}{2} \sum_i |\langle z, u_i \rangle|$. Hence $\Pi L = \sum_i [-u_i/2, u_i/2]$, and the usual volume formula for a zonotope gives $|\Pi L| = \mathcal{H}(L)$. Direct facet parametrization of the unit-height pyramid gives its side pairs and bottom pair as

$$\left(\frac{(u_i, b_i)}{p}, \frac{b_i}{p} \right), \quad ((0, -|L|), 0), \quad (\text{B.2})$$

respectively. One way to check the factor $1/p$ is to integrate the $(p-1)$ -dimensional slices of each side face: its oriented area vector is $(u_i, b_i) \int_0^1 (1-t)^{p-1} dt = (u_i, b_i)/p$. Expanding the corresponding minors gives

$$\mathcal{H}(\text{pyr } L) = p^{-(p+1)} \mathcal{L}(L) + |L| p^{-p} \mathcal{H}(L), \quad (\text{B.3})$$

$$\mathcal{L}(\text{pyr } L) = |L| p^{-(p+1)} \mathcal{L}(L), \quad |\text{pyr } L| = \frac{|L|}{p+1}. \quad (\text{B.4})$$

For the second identity, an all-side lifted minor is zero because its last two coordinate rows coincide; minors using the bottom give the stated expression. Substituting (B.3) into the definition of a therefore proves the exact identity

$$a(L) = \frac{\mathcal{L}(L)}{p|L|\mathcal{H}(L)}. \quad (\text{B.5})$$

These computations concern the original projection bodies, not a substitute invariant. The standard projection covariance $\Pi(AL) = |\det A|A^{-T}\Pi L$ shows that R , and hence a and e , are affine invariant; affine maps of a base extend to its pyramid. The quantity E is affine invariant as well.

B.2 Normalization and the finite cone law

Affinely normalize the prescribed maximum S so that its own centroid is zero and its image contains the closed unit ball. Continue to write K, S for the images. With $n = d + 1$, maximum-volume replacement gives

$$|\alpha_i^S(x)| \leq 1 \quad (x \in K). \quad (\text{B.6})$$

Indeed replacing vertex i by x multiplies the absolute volume by $|\alpha_i^S(x)|$. Since $\sum_i \alpha_i^S(x) = 1$ and the centroid is zero, $-x/d$ has S -coordinates $(1 - \alpha_i^S(x))/d \geq 0$. Thus

$$-K \subset dS \subset dK, \quad E(K, S) \leq n. \quad (\text{B.7})$$

The second conclusion uses the exact centered-dilation description

$$g_S + (1 + \tau)(S - g_S) = \{x : \alpha_i^S(x) \geq -\tau/n \text{ for all } i\}. \quad (\text{B.8})$$

For the actual irredundant facets F of K , use the notation $n_F, s_F, h_F = h_K(n_F)$ and put

$$X_F = \frac{n_F}{h_F}, \quad p_F = \frac{s_F h_F}{d|K|}, \quad \mu = \sum_F p_F \delta_{X_F}. \quad (\text{B.9})$$

Here $h_F \geq 1$ because the unit ball is in K . The cone-volume identity $\sum_F s_F h_F = d|K|$ and the surface-area balance $\sum_F s_F n_F = 0$ show that μ is a centered probability law. Its atoms are precisely actual facet normals divided by their positive support numbers, with $\|X_F\| \leq 1$ and $h_K(X_F) = 1$. Boundedness of K forces these atoms to span $\mathbb{R}^d$; centeredness with positive weights then gives full affine span. In particular the horizontal and lifted moments A, B of Section A are positive.

Cauchy's formula and centeredness give the exact brightness identities

$$m_K(z) := \mathbb{E}(\langle z, X \rangle)_+ = \frac{b_K(z)}{d|K|}, \quad f_K(z) := \mathbb{E}|\langle z, X \rangle| = 2m_K(z). \quad (\text{B.10})$$

Expanding the iid determinants by distinct facet subsets gives

$$A = \frac{d!\mathcal{H}(K)}{(d|K|)^d}, \quad B = \frac{(d+1)!\mathcal{L}(K)}{(d|K|)^{d+1}}.$$

Together with (B.5), this proves

$$a(K) = \frac{B}{nA}, \quad \frac{B-A}{B} = \frac{ne(K)}{1+ne(K)}. \quad (\text{B.11})$$

Jensen's inequality, or (A.2), proves $B \geq A$ and hence $e(K) \geq 0$.

Let $N = h_{K-K}$, a continuous seminorm. At every atom,

$$N(X_F) = h_K(X_F) + h_K(-X_F) \leq 1 + d = n$$

by (B.7). Lemma A.1, with $R_0 = n$, gives $d + 1$ affinely independent *actual atoms* w_i and an assignment with

$$h = \mathbb{E}N(X - w_{I(X)}) \leq \frac{Ce(K)}{1 + ne(K)}, \quad C = n^3(n + 1). \quad (\text{B.12})$$

Thus every selected w_i is an actual facet normal, a fact that is stronger than merely knowing that an anchor belongs to an unspecified support law.

B.3 Width and same-anchor enclosure

For every $z, y \in \mathbb{R}^d$ one has

$$|\langle z, y \rangle| \leq d m_K(z) N(y). \quad (\text{B.13})$$

To see this, first take unit v and $c = \langle y, v \rangle \neq 0$. In coordinates $x = q + tv$ with $q \in v^\perp$, the determinant-one shear $(q, t) \mapsto (q, t + \langle y, q \rangle / c)$ places K in a cylinder with base the projection of K and height $N(y)/|c|$. Thus $|c||K| \leq N(y)\pi_K(v)$. Scaling v to z and using (B.10) proves (B.13); zero cases are immediate.

Let $p_i = \mu\{I = i\}$. Positive-part Lipschitz continuity gives

$$\sum_i p_i (\langle z, w_i \rangle)_+ \geq (1 - dh) m_K(z). \quad (\text{B.14})$$

Assume $h < 1/(2d)$. On the unit sphere, m_K has a positive lower bound, since K contains the unit ball. Hence for every unit z some anchor has positive pairing with z , uniformly bounded away from zero. Support separation puts zero in the interior of $\text{conv}(w_0, \dots, w_d)$. Its polar

$$P = \{x : \langle w_i, x \rangle \leq 1 \text{ for } 0 \leq i \leq d\} \quad (\text{B.15})$$

is therefore a genuine d -simplex containing K . Every defining inequality is an actual K -facet inequality. The corresponding full facet of K lies in a facet of P , providing the actual-facet incidence used in Theorem 1.1.

For $z \in P$, the left side of (B.14) is at most one; therefore $m_K(z) \leq 2$. Equation (B.13) and support separation then give the intrinsic containment

$$P \subset 2d(K - K). \quad (\text{B.16})$$

B.4 Probability correction and relative projection error

Let $\lambda_i > 0$ be the barycentric coordinates of zero in the anchor simplex. Let q_i be the vertex of P opposite the facet with normal w_i . The anchor-simplex coordinates satisfy

$$\alpha_i^w(x) = \lambda_i(1 - \langle q_i, x \rangle). \quad (\text{B.17})$$

Both sides agree at each anchor. The actual cone probabilities of P satisfy the same mass and mean equations as λ_i and hence equal λ_i . Put $c = \sum_i p_i w_i$. Since $\mathbb{E}X = 0$, (B.16) gives $|\langle q_i, c \rangle| \leq 2dh$. Evaluating (B.17) at c yields

$$p_i = \lambda_i(1 - \langle q_i, c \rangle) \geq (1 - 2dh)\lambda_i.$$

Also (B.13) and the absolute-value triangle inequality give

$$\sum_i p_i |\langle z, w_i \rangle| \leq f_K(z) + dm_K(z)h = (1 + dh/2)f_K(z).$$

Write $\theta = 2dh < 1$. The exact brightness formulas for P and K imply

$$(1 - \theta) \frac{\pi_P(u)}{|P|} \leq (1 + \theta/4) \frac{\pi_K(u)}{|K|} \quad (\|u\| = 1). \quad (\text{B.18})$$

The finite first mixed-volume formula and Minkowski's first inequality give

$$\left(\frac{|P|}{|K|} \right)^{1/d} \leq \frac{V(K[d-1], P)}{|K|} = \mathbb{E}h_P(X). \quad (\text{B.19})$$

Indeed the last expression is $\sum_F s_F h_P(n_F)/(d|K|)$. Since $h_P(w_i) = 1$, subadditivity and (B.16) bound it by $1 + 2dh = 1 + \theta$. Combining (B.18) and (B.19),

$$\frac{\pi_K(u)}{\pi_P(u)} \geq \frac{1 - \theta}{(1 + \theta)^d(1 + \theta/4)}.$$

For factors in $[0, 1]$, the deficit of their product is at most the sum of their deficits. Thus the missing fraction above is at most

$$\theta + d\theta + \theta/4 = 2d(d + 5/4)h \leq 3d^2h \quad (d \geq 3). \quad (\text{B.20})$$

Here $1 - (1 + \theta)^{-d} \leq d\theta$ follows from $1 - t^d \leq d(1 - t)$ for $0 \leq t \leq 1$.

Proposition B.1 (Finite actual-facet enclosure). *For $d \geq 3$, if $0 \leq e(K) \leq H_d^{-1}$, the normalized K is contained in a simplex P whose facet hyperplanes are actual K -facet hyperplanes, and*

$$\pi_P(u) - \pi_K(u) \leq H_d e(K) \pi_P(u) \quad (\|u\| = 1). \quad (\text{B.21})$$

Proof. Since $H_d = 3d^2C$, (B.12) gives $h \leq 1/(3d^2) < 1/(2d)$ even at $e(K) = H_d^{-1}$. All strict enclosure gates hold. Equations (B.20) and (B.12) give $3d^2h \leq H_d e(K)$. $\square$

Every step used the same finite actual facet law. No compact approximation, subsequence, or limiting-law identification is needed for this proposition.

C Proof coverage and verification boundaries

C.1 Mathematical provenance

The manuscript organizes the research series dated 8 October 2026, whose frozen source archive has SHA-256

631a80b594fb9a597fef15e1c76870956ec
b768fbbb14f3b7953e1809970296e.

The companion source map identifies each statement with its mathematical source and a version-bound review. The 55 source-manifest payload files are preserved byte for byte; the source archive also contains its manifest. This manuscript is a new exposition and does not overwrite any frozen source or earlier manuscript.

Sections 2–4 reproduce the complete four-note few-facet series: rooted-forest determinants and equality, arbitrary nonvertex maxima, original-centroid excess, arbitrary one-corner bodies, actual signed-cofactor identities, and the full dimensional squeeze. The lower-body

facet certificate comes from the retained realized-body obstruction note. The exact truncation identity is included rather than using an unidentified geometric surrogate for the defect.

Section 5 includes the elementary and Brunn–Minkowski core proofs, the finite-spectrum refinement and its uniform branch, the beta/binomial face hierarchy, and the necessary countermodel tests. Section 6 includes weighted cycles, attached trees, the replacement-minor reduction, averaging for every prescribed maximum, the compatible actual-defect consequence, and the additional order-three, positive-definite, and local-basin statements. The latter finite structures do not establish unrestricted retention.

Appendices A and B reproduce the finite determinant-assignment and actual-facet enclosure proofs from the existing vertex-excess manuscript exposition, adapted only to this paper’s references. They expand the exact finite-law interface identified in the frozen series. In particular, facet incidence is proved by selection from actual irredundant facets; it is not claimed as an unproved consequence of a compact-limit formal theorem. Classical mixed-volume and projection formulas are explicit standard inputs, not newly proved stability results.

C.2 The traditional review record

The four-note few-facet series received a separate independent traditional model review with a PASS disposition bound to the frozen versions. That review checked all four proofs, the arbitrary prescribed- S quantifier, the original invariant, the finite-facet interface, and the realized lower examples. It inspected the retained interface without re-proving the entire upstream actual-body theory. Its only source edit was the resolved replacement of “span” by “form a basis” in the rooted-forest rank argument. This manuscript follows the corrected version.

The full-projection series received the parent research worker’s traditional proof and integration review. The uniform $\rho \leq 1/3$ branch was contributed by that parent and cross-checked by the originating projection worker. Thus the parent review is not independent review of its own authored refinement, and the entire series is not presented as having a uniformly independent authorship-and-review arrangement.

The inverse-minor series received a parent traditional reading and a separate theoretical worker’s proof work and focused integration audit. The latter’s final scope includes the all-body coordinate control, replacement-minor reduction, every-maximizer averaging, compact one-skeleton reduction, conditional actual-defect consequence, and the order-three and golden-ratio formulas. The corrected hypothesis uses *every extreme point* on the admissible simplex’s one-skeleton; an earlier ambiguous wording is not retained. The separate focused audit treats the upstream actual enclosure chain as an input. Because the worker also contributed proof material, it is not described here as a wholly external independent review of every matrix proof.

All of these are traditional model-assisted mathematical checks, not professional peer review. They bind source versions. They do not by themselves certify the typesetting, cross-references, or transcription of this new manuscript. Its build, page-by-page visual inspection, proof coverage, and source hashes are recorded in the companion candidate QA report; final independent copy review remains a separate gate.

C.3 Exact finite checks

The source review record contains the following reproducible checks.

- The clipping checker covers 12 rational examples and all 12,769 vertex tuples. An independent Bareiss implementation verifies their full forest product and equality classification:

8,050 singular tuples, 4,719 rooted forests, 64 maxima, and 4,655 strict nonmaximum forests.

- The author cofactor checker covers 60 balanced rational matrices and 32 actual-truncation comparisons. A separate integer checker covers 210 augmented matrices and 4,830 omitted-pair identities. General matrix samples test algebra, not geometric realizability as facet data.
- Projection checks cover 189 exact beta/binomial identities and 780 powered rational comparisons for the uniform spectrum branch in $2 \leq m \leq 40$. The all-dimensional conclusion uses the written scalar proof, not these finite samples.
- Inverse-minor replay covers 140 order-three and 287 support-two rational matrices, including 149 near-endpoint cases, the symbolic golden-ratio identities, averaging and cap constants, and the known gaps in two previously proposed false unrestricted two-minor bounds.

The author’s four certificate outputs were replayed byte identically in the frozen series. Exact finite tests support arithmetic and implementation consistency; they do not replace the universal proofs or certify novelty.

C.4 Formal status and the unresolved question

The earlier unrestricted quadratic-dimensional result has a retained strict Lean compilation and same-implementation kernel-replay record. No new Lean compilation, kernel replay, or formal bridge for the present few-facet, clipping, cofactor, full-projection, or support-two results has been performed. Reusing an earlier compiled theorem does not formalize these new traditional arguments. No external independent kernel-checker implementation is claimed.

The remaining unrestricted target is a dimension-free constant C such that $E(K, S) \leq Cn\rho$ follows from the full actual projection information for every prescribed maximum S , or a realized actual-body family proving superlinear dimension growth. The core provides a structured comparison matrix $(1 - t)I + (t/n)J$, but its simplex need not be maximum. Face-depth inequalities are necessary and are not sufficient for projection closeness. Support on three or more coordinates does not reduce to the unicyclic argument. The golden-ratio example obstructs small global constants but is not a $c \downarrow 1$ counterexample. No arbitrary determinant model is identified with the actual geometric defect without a separate calculation.

References

- [1] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications 151, Cambridge University Press, 2014. <https://doi.org/10.1017/CB09781139003858>.
- [2] Entry 005 research project, *A dimension-stability structural series*, 8 October 2026. Unpublished source notes and version-bound traditional reviews. Mathematical notes, certificate scripts, provenance, and exact source hashes are identified in the accompanying source map. No priority certification is asserted.
- [3] Entry 005 research project, *Quadratic-dimensional simplex stability*, 7 October 2026. Retained actual-body proof report and formal verification record. The finite actual-facet specialization needed here is reproduced in Appendices A–B.